

Shiv Chhatrapati Shikshan Sanstha's

Rajarshi Shahu Mahavidyalaya, Latur

(Autonomous)

Department of Information Technology

A project report on

“Browser Based Phishing Detection System(Extension)”

Submitted By

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Exam Seat No: **RDS2662615**

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&

Of

B.Sc. CS III Year (VI-Sem)

Under the Guidance of

Prof. Riyaj Shaikh

Academic Year

2025-2026

Date: / /2026

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CERTIFICATE

This is to certify that the Project Report on

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B.Sc. CS III Year (VI-Sem)

As per the requirements of **Swami Ramanand Teerth**

Marathwada University, Nanded in partial fulfilment of Degree

“Bachelor of Computer Science”

For the Academic Year 2025-2026

Guide
Prof. Riyaj Shaikh

Project Incharge
Prof. A. K .Shaikh

Examiner

Head of Department
Prof. Jyoti V. Mashalkar

Acknowledgement

I would like to express my heartfelt gratitude to all those who contributed to the successful completion of this project, “**Browser Based Phishing Detection System (Extension)**”. I am sincerely thankful to my project guide, **Prof. Riyaj Shaikh** , Project Incharge **Prof. A. K. Shaikh** and respected **Dr. Mahadev Gavhane Sir** for his unwavering support, invaluable guidance, and continuous encouragement throughout the project's lifecycle.

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Abstract

Phishing attacks are one of the most common cybersecurity threats, where attackers attempt to steal sensitive information such as usernames, passwords, and banking details by creating fake websites that appear legitimate. This project presents a “**Browser-Based Phishing Detection System(Extension)**” implemented as an extension that helps users identify and avoid malicious websites in real time. The extension works by analyzing various features of a webpage, including URL structure, domain age, presence of HTTPS, suspicious keywords, and page behavior. It uses predefined rules and/or machine learning techniques to classify websites as safe or potentially harmful. When a user visits a suspicious website, the extension immediately alerts them through warning messages, helping prevent data theft. This system operates directly within the browser, ensuring fast detection without requiring external software. It is lightweight, user-friendly, and enhances online security by providing instant feedback. The proposed solution aims to reduce phishing risks and improve user awareness, making web browsing safer for individuals and organizations.

Introduction

Phishing is a type of cyberattack where attackers create fake websites or emails to trick users into revealing sensitive information such as passwords, credit card details, and personal data. With the increasing use of the internet for banking, shopping, and communication, phishing attacks have become more sophisticated and dangerous. To address this issue, this project proposes a “**Browser-Based Phishing Detection System(Extension)**” implemented as a browser extension. The system helps users identify phishing websites in real time and provides alerts to prevent data theft. The system uses rule-based techniques and can be enhanced with machine learning algorithms to classify websites as safe or suspicious. When a user visits a potentially harmful website, the extension immediately alerts them, helping to prevent data theft and cyber fraud. Since the system operates directly within the browser, it provides fast and efficient detection without requiring additional software installation.

Overall, this project aims to enhance online security, improve user awareness, and provide a practical solution to reduce the risks associated with phishing attacks.

Objectives

Real-time detection: The extension checks websites instantly to protect users before they enter data.

URL analysis: It identifies suspicious patterns like long URLs, special characters, and fake domains.

Website feature checking: It verifies HTTPS, SSL certificates, and domain details to judge safety.

Classification: The system labels websites as safe or phishing using rules or machine learning.

User alerts: It gives quick warnings when a suspicious site is detected.

User-friendly design: The extension is simple and easy for anyone to use.

User awareness: It helps users understand phishing and stay cautious online.

Data protection: It reduces the risk of losing sensitive information.

Fast performance: It works quickly without slowing down the browser.

Implementation

The Browser-Based Phishing Detection System is developed using a combination of frontend, backend, and database technologies to ensure efficient real-time detection.

Frontend technologies: HTML, CSS, JavaScript

The user interface is built using HTML, CSS, and JavaScript.

- HTML is used to structure the extension popup and UI components.
- CSS is used for styling and creating a user-friendly design.
- JavaScript handles logic, user interactions, and communication with the backend.
- The Chrome Extension API is used to capture URLs, interact with browser tabs, and display alerts to users.

Backend technologies: Python(Flask),Node.js(Express)

- The backend is responsible for processing data and detecting phishing websites.
- **Python (Flask):**
Used to build REST APIs that analyze URLs and webpage features. It may also run machine learning models for classification.
- **Node.js (Express):**
- Handles server-side operations such as request handling, routing, and communication between frontend and database.

Database: MongoDB

MongoDB is used to store:

- Phishing and legitimate URL datasets
- User activity logs (optional)
- Detection results

MongoDB provides flexibility and fast data retrieval, making it suitable for real-time applications.

Folder Structure:

phishing-detection-system/

├── extension/ # Chrome Extension (Frontend)

| ├── manifest.json # Extension configuration

```
| | └─ popup.html      # UI popup
| | └─ popup.css       # Styling
| | └─ popup.js        # UI logic
| | └─ background.js   # Background script
| | └─ content.js      # Runs on web pages
| | └─ icons/          # Extension icons
| |   └─ icon16.png
|   └─ utils/          # Helper functions
|     └─ api.js        # API calls to backend
└─ backend/            # Flask Backend (Rule-based)
    └─ app.py          # Main Flask app
    └─ routes/         # API routes
        └─ detect.py   # Phishing detection API
    └─ utils/          # Helper functions
        └─ feature_extraction.py
    └─ requirements.txt # Python dependencies
└─ database/           # MongoDB setup
    └─ mongodb_config.js # DB connection
```

Source Code

****Front-end****

1. Dashboard.html:

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Dashboard — PhishingGuard</title>

<link rel="preconnect" href="https://fonts.googleapis.com">

<link
href="https://fonts.googleapis.com/css2?family=Space+Mono:wght@400;700&family=Syne:wght@400;600;700;800&display=swap" rel="stylesheet">

<script src="https://cdnjs.cloudflare.com/ajax/libs/Chart.js/4.4.1/chart.umd.min.js"></script>

<style>

:root {

  --bg:#0a0e1a; --panel:#111827; --panel2:#0f1929; --border:#1e2d45;

  --accent:#00d4ff; --accent2:#ff3c6e; --green:#00ff88; --yellow:#ffd60a;
```

```
--text:#e2e8f0; --muted:#64748b; --sidebar-w:240px; }

* { margin:0; padding:0; box-sizing:border-box; }

body { background:var(--bg); font-family:'Syne',sans-serif; color:var(--text); display:flex; min-height:100vh; }

/* Sidebar */

.sidebar {

    width:var(--sidebar-w); background:var(--panel); border-right:1px solid var(--border);

    display:flex; flex-direction:column; position:fixed; height:100vh; z-index:100; }

.sidebar-logo { padding:28px 24px; border-bottom:1px solid var(--border); }

.sidebar-logo .name { font-size:18px; font-weight:800; }

.sidebar-logo .name span { color:var(--accent); }

.sidebar-logo .tagline { font-size:11px; color:var(--muted); margin-top:2px; }

.nav { flex:1; padding:20px 12px; }

.nav-item {

    display:flex; align-items:center; gap:12px; padding:11px 14px;

    border-radius:8px; font-size:14px; font-weight:600; cursor:pointer;

    color:var(--muted); transition:all .2s; text-decoration:none; margin-bottom:4px;}
```

```
.nav-item:hover { background:rgba(0,212,255,0.07); color:var(--text); }

.nav-item.active { background:rgba(0,212,255,0.12); color:var(--accent); }

.nav-item .icon { font-size:18px; width:22px; text-align:center; }

.sidebar-user { padding:16px 20px; border-top:1px solid var(--border); }

.user-badge { display:flex; align-items:center; gap:10px; }

.user-avatar {

    width:36px; height:36px; border-radius:50%;

    background:linear-gradient(135deg, var(--accent), var(--accent2));

    display:flex; align-items:center; justify-content:center;

    font-weight:700; font-size:14px; color:#000; flex-shrink:0; }

.user-name { font-size:13px; font-weight:600; }

.user-email { font-size:11px; color:var(--muted); }

.logout-btn { margin-top:12px; width:100%; padding:8px; background:transparent; border:1px
solid var(--border); border-radius:6px; color:var(--muted); font-size:12px; font-
family:'Syne',sans-serif; cursor:pointer; transition:all .2s; }

.logout-btn:hover { border-color:var(--accent2); color:var(--accent2); }

/* Main content */ .main { margin-left:var(--sidebar-w); flex:1; padding:32px; overflow-y:auto;
}
```

```
.page-header { margin-bottom:32px; }
```

```
.page-header h1 { font-size:28px; font-weight:800; letter-spacing:-0.5px; }
```

```
.page-header p { color:var(--muted); font-size:14px; margin-top:4px; }
```

```
/* Stat cards */
```

```
.stats-grid { display:grid; grid-template-columns:repeat(4,1fr); gap:16px; margin-bottom:28px; }
```

```
.stat-card {
```

```
background:var(--panel); border:1px solid var(--border); border-radius:12px;
```

```
padding:20px; position:relative; overflow:hidden; transition:transform .2s; }
```

```
.stat-card:hover { transform:translateY(-2px); }
```

```
.stat-card::before {
```

```
content:""; position:absolute; top:0; left:0; right:0; height:3px; }
```

```
.stat-card.blue::before { background:var(--accent); }
```

```
.stat-card.red::before { background:var(--accent2); }
```

```
.stat-card.green::before { background:var(--green); }
```

```
.stat-card.yellow::before { background:var(--yellow);}
```

```
.stat-icon { font-size:28px; margin-bottom:10px; }
```

```
.stat-num { font-size:32px; font-weight:800; letter-spacing:-1px; font-family:'Space Mono',monospace; }
```

```
.stat-label { font-size:12px; color:var(--muted); margin-top:4px; text-transform:uppercase; letter-spacing:0.5px; }
```

```
.stat-card.blue .stat-num { color:var(--accent); }
```

```
.stat-card.red .stat-num { color:var(--accent2); }
```

```
.stat-card.green .stat-num { color:var(--green); }
```

```
.stat-card.yellow .stat-num { color:var(--yellow); }
```

```
/* Charts row */ .charts-row { display:grid; grid-template-columns:2fr 1fr; gap:20px; margin-bottom:28px; }
```

```
.chart-card {
```

```
background:var(--panel); border:1px solid var(--border); border-radius:12px; padding:24px; }
```

```
.card-title { font-size:14px; font-weight:700; color:var(--muted); text-transform:uppercase; letter-spacing:1px; margin-bottom:20px; }
```

```
/* Recent scans */
```

```
.scans-card { background:var(--panel); border:1px solid var(--border); border-radius:12px; padding:24px; }
```

```
.scan-row {
```

```
display:flex; align-items:center; gap:12px;
```

```
padding:12px 0; border-bottom:1px solid var(--border); }

.scan-row:last-child { border-bottom:none; }

.scan-badge {

padding:3px 10px; border-radius:4px; font-size:10px; font-weight:700;

text-transform:uppercase; letter-spacing:0.5px; flex-shrink:0; }

.badge-safe { background:rgba(0,255,136,0.15); color:var(--green); }

.badge-medium { background:rgba(255,214,10,0.15); color:var(--yellow); }

.badge-high { background:rgba(255,107,0,0.15); color:#ff6b00; }

.badge-critical { background:rgba(255,60,110,0.15); color:var(--accent2); }

.scan-url { flex:1; font-family:'Space Mono',monospace; font-size:12px; color:var(--text);
overflow:hidden; text-overflow:ellipsis; white-space:nowrap; }

.scan-time { font-size:11px; color:var(--muted); flex-shrink:0; }

.scan-count { font-size:12px; color:var(--muted); flex-shrink:0; }

.empty-state { text-align:center; padding:40px; color:var(--muted); }

.empty-state .icon { font-size:48px; margin-bottom:12px; }

.loading { text-align:center; padding:20px; color:var(--muted); font-size:13px; }

</style></head>
```

<body>

<nav class="sidebar">

<div class="sidebar-logo">

<div class="name">PhishingGuard</div>

<div class="tagline">Real-time threat protection</div> </div>

<div class="nav">


Dashboard

 Scan History

 Profile </div>

<div class="sidebar-user">

<div class="user-badge">

<div class="user-avatar" id="userInitial">?</div> <div>

<div class="user-name" id="userName">Loading...</div>

<div class="user-email" id="userEmail"></div> </div> </div>

<button class="logout-btn" onclick="logout()">Sign Out</button>

</div> </nav>

```
<main class="main">
```

```
<div class="page-header">
```

```
<h1>Security Dashboard</h1>
```

```
<p>Overview of your browsing protection activity</p> </div>
```

```
<div class="stats-grid">
```

```
<div class="stat-card blue">
```

```
<div class="stat-icon"> </div>
```

```
<div class="stat-num" id="statTotal">—</div>
```

```
<div class="stat-label">Total Scans</div> </div>
```

```
<div class="stat-card red">
```

```
<div class="stat-icon"> </div>
```

```
<div class="stat-num" id="statThreats">—</div>
```

```
<div class="stat-label">Threats Detected</div> </div>
```

```
<div class="stat-card green">
```

```
<div class="stat-icon"> </div>
```

```
<div class="stat-num" id="statSafe">—</div>
```

<div class="stat-label">Safe Sites</div></div>

<div class="stat-card yellow">

<div class="stat-icon"></div>

<div class="stat-num" id="statRate">—</div>

<div class="stat-label">Protection Rate</div> </div> </div>

<div class="charts-row">

<div class="chart-card">

<div class="card-title">☒ Last 7 Days Activity</div>

<canvas id="activityChart" height="110"></canvas> </div>

<div class="chart-card">

<div class="card-title"> Threat Breakdown</div>

<canvas id="donutChart" height="140"></canvas> </div> </div>

<div class="scans-card">

<div class="card-title"> Recent Scans</div>

<div id="recentScans"><div class="loading">Loading recent scans...</div></div>

</div> </main>

```
<script>
```

```
const API = 'http://localhost:5000';
```

```
const token = localStorage.getItem('pd_token');
```

```
if (!token) window.location.href = '/login';
```

```
const user = JSON.parse(localStorage.getItem('pd_user') || '{}');
```

```
document.getElementById('userName').textContent = user.name || '—';
```

```
document.getElementById('userEmail').textContent = user.email || '';
```

```
document.getElementById('userInitial').textContent = (user.name || 'U')[0].toUpperCase();
```

```
function logout() {
```

```
    localStorage.removeItem('pd_token');
```

```
    localStorage.removeItem('pd_user');
```

```
    window.location.href = '/login'; }
```

```
async function apiFetch(path) {
```

```
    const res = await fetch(API + path, { headers: { 'Authorization': 'Bearer ' + token } });
```

```
    if (res.status === 401) { logout(); return null; }
```

```
    return res.json(); }
```

```
async function loadDashboard() {

  const [stats, scans] = await Promise.all([

    apiFetch('/api/scans/stats'),

    apiFetch('/api/scans?limit=8')    ]);

  if (stats) {

    document.getElementById('statTotal').textContent = stats.total_scans;

    document.getElementById('statThreats').textContent = stats.threats_detected;

    document.getElementById('statSafe').textContent = stats.safe_sites;

    const rate = stats.total_scans > 0

      ? Math.round((stats.safe_sites / stats.total_scans) * 100) + '%' : '—';

    document.getElementById('statRate').textContent = rate;

    renderCharts(stats);    }

  if (scans) renderRecentScans(scans.scans);    }

function renderCharts(stats) {

  const activity = stats.recent_activity || [];

  const labels = activity.map(d => d._id.slice(5)); // MM-DD
```

```
const scanData = activity.map(d => d.scans);

const threatData = activity.map(d => d.threats);

new Chart(document.getElementById('activityChart'), {

  type: 'line',

  data: {

    labels,

    datasets: [

      { label: 'Scans', data: scanData, borderColor: '#00d4ff', backgroundColor:
'rgba(0,212,255,0.08)', tension: 0.4, fill: true },

      { label: 'Threats', data: threatData, borderColor: '#ff3c6e', backgroundColor:
'rgba(255,60,110,0.08)', tension: 0.4, fill: true } ] },

  options: {

    responsive: true,

    plugins: { legend: { labels: { color: '#64748b', font: { family: 'Syne' } } } },

    scales: {

      x: { ticks: { color: '#64748b' }, grid: { color: '#1e2d45' } },

      y: { ticks: { color: '#64748b' }, grid: { color: '#1e2d45' }, beginAtZero: true } } } });

// Donut chart
```

```
const bd = stats.breakdown || {};  
  
new Chart(document.getElementById('donutChart'), {  
  
  type: 'doughnut',  
  
  data: {  
  
    labels: ['Safe', 'Medium', 'High', 'Critical'],  
  
    datasets: [{  
  
      data: [bd.safe||0, bd.medium||0, bd.high||0, bd.critical||0],  
  
      backgroundColor: ['#00ff88','#ffd60a','#ff6b00','#ff3c6e'],  
  
      borderWidth: 0,  
  
      hoverOffset: 6  
  
    }]  
  },  
  
  options: {  
  
    cutout: '68%',  
  
    plugins: { legend: { position:'bottom', labels: { color:'#64748b', padding:12, font:{  
family:'Syne', size:12 } } } }  
  
  }  
});  
  
function renderRecentScans(scans) {
```

```
const el = document.getElementById('recentScans');
```

```
if (!scans || scans.length === 0) {
```

```
    el.innerHTML = '<div class="empty-state"><div class="icon">
```

```
    return; }
```

```
    el.innerHTML = scans.map(s => {
```

```
        const badgeClass = 'badge-' + (s.status === 'safe' ? 'safe' : s.status === 'medium' ? 'medium' : s.status === 'high' ? 'high' : 'critical');
```

```
        const time = new Date(s.scanned_at).toLocaleString();
```

```
        let hostname = s.url;
```

```
        try { hostname = new URL(s.url).hostname; } catch(e) {}
```

```
        return `
```

```
<div class="scan-row">
```

```
    <span class="scan-badge ${badgeClass}">${s.status}</span>
```

```
    <span class="scan-url" title="${s.url}">${hostname}</span>
```

```
    <span class="scan-count">${s.threat_count} issue${s.threat_count !== 1 ? 's' : ''}</span>
```

```
    <span class="scan-time">${time}</span>
```

```
        </div>`;

    }).join("");  }

loadDashboard();

</script></body></html>
```

2.History.html

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Scan History — PhishingGuard</title>

<link rel="preconnect" href="https://fonts.googleapis.com">

<link
href="https://fonts.googleapis.com/css2?family=Space+Mono:wght@400;700&family=Syne:wght@400;600;700;800&display=swap" rel="stylesheet">

<style>

    :root { --bg:#0a0e1a; --panel:#111827; --border:#1e2d45; --accent:#00d4ff; --accent2:#ff3c6e;
--green:#00ff88; --yellow:#ffd60a; --text:#e2e8f0; --muted:#64748b; --sidebar-w:240px; }
```

```
* { margin:0; padding:0; box-sizing:border-box; }
```

```
body { background:var(--bg); font-family:'Syne',sans-serif; color:var(--text); display:flex; min-height:100vh; }
```

```
.sidebar { width:var(--sidebar-w); background:var(--panel); border-right:1px solid var(--border); display:flex; flex-direction:column; position:fixed; height:100vh; z-index:100; }
```

```
.sidebar-logo { padding:28px 24px; border-bottom:1px solid var(--border); }
```

```
.sidebar-logo .name { font-size:18px; font-weight:800; }
```

```
.sidebar-logo .name span { color:var(--accent); }
```

```
.sidebar-logo .tagline { font-size:11px; color:var(--muted); margin-top:2px; }
```

```
.nav { flex:1; padding:20px 12px; }
```

```
.nav-item { display:flex; align-items:center; gap:12px; padding:11px 14px; border-radius:8px; font-size:14px; font-weight:600; cursor:pointer; color:var(--muted); transition:all .2s; text-decoration:none; margin-bottom:4px; }
```

```
.nav-item:hover { background:rgba(0,212,255,0.07); color:var(--text); }
```

```
.nav-item.active { background:rgba(0,212,255,0.12); color:var(--accent); }
```

```
.nav-item .icon { font-size:18px; width:22px; text-align:center; }
```

```
.sidebar-user { padding:16px 20px; border-top:1px solid var(--border); }
```

```
.user-badge { display:flex; align-items:center; gap:10px; }
```

```
.user-avatar { width:36px; height:36px; border-radius:50%; background:linear-gradient(135deg, var(--accent), var(--accent2)); display:flex; align-items:center; justify-content:center; font-weight:700; font-size:14px; color:#000; flex-shrink:0; }
```

```
.user-name { font-size:13px; font-weight:600; }
```

```
.user-email { font-size:11px; color:var(--muted); }
```

```
.logout-btn { margin-top:12px; width:100%; padding:8px; background:transparent; border:1px solid var(--border); border-radius:6px; color:var(--muted); font-size:12px; font-family:'Syne',sans-serif; cursor:pointer; transition:all .2s; }
```

```
.logout-btn:hover { border-color:var(--accent2); color:var(--accent2); }
```

```
.main { margin-left:var(--sidebar-w); flex:1; padding:32px; }
```

```
.page-header { margin-bottom:28px; display:flex; align-items:center; justify-content:space-between; }
```

```
.page-header h1 { font-size:28px; font-weight:800; letter-spacing:-0.5px; }
```

```
.page-header p { color:var(--muted); font-size:14px; margin-top:4px; }
```

```
.filters { display:flex; gap:12px; margin-bottom:20px; flex-wrap:wrap; }
```

```
.filter-btn { padding:8px 16px; border-radius:20px; border:1px solid var(--border); background:transparent; color:var(--muted); font-family:'Syne',sans-serif; font-size:13px; cursor:pointer; transition:all .2s; }
```

```
.filter-btn:hover, .filter-btn.active { border-color:var(--accent); color:var(--accent); background:rgba(0,212,255,0.08); }
```

```
.filter-btn.active-red { border-color:var(--accent2); color:var(--accent2); background:rgba(255,60,110,0.08); }
```

```
.history-table { background:var(--panel); border:1px solid var(--border); border-radius:12px;
overflow:hidden; }
```

```
.table-header { display:grid; grid-template-columns:100px 1fr 80px 80px 160px 50px; gap:16px;
padding:14px 20px; border-bottom:1px solid var(--border); font-size:11px; font-weight:700;
color:var(--muted); text-transform:uppercase; letter-spacing:1px; }
```

```
.table-row { display:grid; grid-template-columns:100px 1fr 80px 80px 160px 50px; gap:16px;
padding:14px 20px; border-bottom:1px solid var(--border); align-items:center;
transition:background .15s; }
```

```
.table-row:last-child { border-bottom:none; }
```

```
.table-row:hover { background:rgba(0,212,255,0.03); }
```

```
.badge { padding:4px 10px; border-radius:4px; font-size:10px; font-weight:700; text-
transform:uppercase; letter-spacing:0.5px; display:inline-block; }
```

```
.badge-safe { background:rgba(0,255,136,0.12); color:var(--green); }
```

```
.badge-medium { background:rgba(255,214,10,0.12); color:var(--yellow); }
```

```
.badge-high { background:rgba(255,107,0,0.12); color:#ff6b00; }
```

```
.badge-critical { background:rgba(255,60,110,0.12); color:var(--accent2); }
```

```
.url-cell { font-family:'Space Mono',monospace; font-size:12px; overflow:hidden; text-
overflow:ellipsis; white-space:nowrap; }
```

```
.count-cell { font-family:'Space Mono',monospace; font-size:13px; text-align:center; }
```

```
.time-cell { font-size:12px; color:var(--muted); }
```

```
.del-btn { background:transparent; border:none; color:var(--muted); cursor:pointer; font-size:16px; padding:4px; border-radius:4px; transition:color .2s; } .del-btn:hover { color:var(--accent2); }
```

```
.pagination { display:flex; gap:8px; justify-content:center; margin-top:20px; align-items:center; }
```

```
.page-btn { padding:8px 14px; border-radius:6px; border:1px solid var(--border); background:transparent; color:var(--muted); font-family:'Syne',sans-serif; font-size:13px; cursor:pointer; transition:all .2s; }
```

```
.page-btn:hover, .page-btn.active { border-color:var(--accent); color:var(--accent); }
```

```
.page-btn:disabled { opacity:.3; cursor:not-allowed; }
```

```
.empty-state { text-align:center; padding:60px 20px; color:var(--muted); }
```

```
.empty-state .icon { font-size:48px; margin-bottom:12px; }
```

```
.loading { text-align:center; padding:40px; color:var(--muted); }
```

```
</style></head>
```

```
<body>
```

```
<nav class="sidebar">
```

```
<div class="sidebar-logo">
```

```
<div class="name">Phishing<span>Guard</span></div>
```

```
<div class="tagline">Real-time threat protection</div> </div>
```

```
<div class="nav">
```







<div class="sidebar-user">

<div class="user-badge">

<div class="user-avatar" id="userInitial">?</div>

<div>

<div class="user-name" id="userName">Loading...</div>

<div class="user-email" id="userEmail"></div>

</div> </div>

<button class="logout-btn" onclick="logout()">Sign Out</button> </div> </nav>

<main class="main">

<div class="page-header">

<div>

<h1>Scan History</h1>

<p>All URLs scanned by your extension</p>

</div> </div>

```
<div class="filters">
```

```
<button class="filter-btn active" onclick="setFilter('all',this)">All</button>
```

```
<button class="filter-btn" onclick="setFilter('safe',this)">✔ Safe</button>
```

```
<button class="filter-btn" onclick="setFilter('medium',this)">⚠ Medium</button>
```

```
<button class="filter-btn" onclick="setFilter('high',this)">⬢ High</button>
```

```
<button class="filter-btn" onclick="setFilter('critical',this)">🚨 Critical</button>
```

```
</div>
```

```
<div class="history-table">
```

```
<div class="table-header">
```

```
<div>Status</div>
```

```
<div>URL</div>
```

```
<div style="text-align:center">Threats</div>
```

```
<div></div>
```

```
<div>Time</div>
```

```
<div></div> </div>
```

```
<div id="scanList"><div class="loading">Loading...</div></div> </div>
```

```
<div class="pagination" id="pagination"></div>
```

</main>

<script>

```
const API = 'http://localhost:5000';
```

```
const token = localStorage.getItem('pd_token');
```

```
if (!token) window.location.href = '/login';
```

```
const user = JSON.parse(localStorage.getItem('pd_user') || '{}');
```

```
document.getElementById('userName').textContent = user.name || '—';
```

```
document.getElementById('userEmail').textContent = user.email || '';
```

```
document.getElementById('userInitial').textContent = (user.name || 'U')[0].toUpperCase();
```

```
let currentPage = 1, currentFilter = 'all', totalPages = 1;
```

```
function logout() {
```

```
  localStorage.removeItem('pd_token'); localStorage.removeItem('pd_user');
```

```
  window.location.href = '/login'; }
```

```
function setFilter(f, el) {
```

```
  currentFilter = f; currentPage = 1;
```

```
  document.querySelectorAll('.filter-btn').forEach(b => b.classList.remove('active'));
```

```
  el.classList.add('active');
```

```
loadScans(); }

async function loadScans() {

  const el = document.getElementById('scanList');

  el.innerHTML = '<div class="loading">Loading...</div>';

  let url = `${API}/api/scans?page=${currentPage}&limit=15`;

  const res = await fetch(url, { headers: { 'Authorization': 'Bearer ' + token } });

  if (res.status === 401) { logout(); return; }

  const data = await res.json();

  let scans = data.scans || [];

  if (currentFilter !== 'all') scans = scans.filter(s => s.status === currentFilter);

  totalPages = data.pages;

  if (scans.length === 0) {

    el.innerHTML = '<div class="empty-state"><div class="icon">

---


```

```
let hostname = s.url;

try { hostname = new URL(s.url).hostname; } catch(e) {}

const time = new Date(s.scanned_at).toLocaleString();

return `

<div class="table-row" id="row-${s._id}">

  <div><span class="badge badge-${s.status}">${s.status}</span></div>

  <div class="url-cell" title="${s.url}">${hostname}</div>

  <div class="count-cell">${s.threat_count}</div>

  <div></div>

  <div class="time-cell">${time}</div>

  <div><button class="del-btn" onclick="deleteScan('${s._id}')"
title="Delete"><img alt="trash icon" data-bbox="270 610 285 625"/></button></div>

</div>`;

}).join("");

renderPagination(); }

function renderPagination() {

  const el = document.getElementById('pagination');
```

```

    if (totalPages <= 1) { el.innerHTML = ""; return; }

    let html = `<button class="page-btn" onclick="goPage(${currentPage-1})"
    ${currentPage===1?'disabled':''}>< Prev</button>`;

    for (let i = 1; i <= totalPages; i++) {

        html += `<button class="page-btn ${i===currentPage?'active':''}"
        onclick="goPage(${i})">${i}</button>`;    }

    html += `<button class="page-btn" onclick="goPage(${currentPage+1})"
    ${currentPage===totalPages?'disabled':''}>Next →</button>`;

    el.innerHTML = html;    }

function goPage(p) { if (p < 1 || p > totalPages) return; currentPage = p; loadScans(); }

async function deleteScan(id) {

    if (!confirm('Delete this scan record?')) return;

    const res = await fetch(`${API}/api/scans/${id}`, { method:'DELETE',
    headers:{'Authorization':'Bearer '+token} });

    if (res.ok) { document.getElementById('row-' + id)?.remove(); }    }

    loadScans();

</script></body></html>

```

3. Login.html:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Login — Phishing Detector</title>
```

```
<link rel="preconnect" href="https://fonts.googleapis.com">
```

```
<link  
href="https://fonts.googleapis.com/css2?family=Space+Mono:wght@400;700&family=Syne:wght@  
400;600;800&display=swap" rel="stylesheet">
```

```
<style>
```

```
:root {
```

```
  --bg: #0a0e1a;
```

```
  --panel: #111827;
```

```
  --border: #1e2d45;
```

```
  --accent: #00d4ff;
```

```
  --accent2: #ff3c6e;
```

```
  --green: #00ff88;
```

--text: #e2e8f0;

--muted: #64748b;

--danger: #ff3c6e; }

* { margin:0; padding:0; box-sizing:border-box; }

body {

background: var(--bg);

font-family: 'Syne', sans-serif;

color: var(--text);

min-height: 100vh;

display: flex;

align-items: center;

justify-content: center;

overflow: hidden;

body::before {

content: '';

position: fixed; inset: 0;

background-image:

linear-gradient(rgba(0,212,255,0.04) 1px, transparent 1px),

linear-gradient(90deg, rgba(0,212,255,0.04) 1px, transparent 1px);

background-size: 40px 40px;

pointer-events: none;

z-index: 0; }

.glow-orb {

position: fixed;

border-radius: 50%;

filter: blur(80px);

pointer-events: none;

z-index: 0; }

.orb1 { width:400px; height:400px; background:rgba(0,212,255,0.07); top:-100px; left:-100px; }

.orb2 { width:300px; height:300px; background:rgba(255,60,110,0.07); bottom:-80px; right:-80px;
}

.card {

position: relative; z-index: 1;

background: var(--panel);

border: 1px solid var(--border);

border-radius: 16px;

padding: 48px 44px;

width: 420px;

box-shadow: 0 0 60px rgba(0,0,0,0.5), 0 0 0 1px rgba(0,212,255,0.05); }

.logo {

display: flex; align-items: center; gap: 12px;

margin-bottom: 36px; }

.logo-icon {

width: 44px; height: 44px;

background: linear-gradient(135deg, var(--accent), var(--accent2));

border-radius: 10px;

display: flex; align-items: center; justify-content: center;

font-size: 22px; }

.logo-text { font-size: 20px; font-weight: 800; letter-spacing: -0.5px; }

```
.logo-text span { color: var(--accent); }
```

```
h2 { font-size: 26px; font-weight: 800; margin-bottom: 6px; }
```

```
.subtitle { color: var(--muted); font-size: 14px; margin-bottom: 32px; }
```

```
.field { margin-bottom: 20px; }
```

```
label { display: block; font-size: 12px; font-weight: 600; color: var(--muted); text-transform: uppercase; letter-spacing: 1px; margin-bottom: 8px; }
```

```
input {
```

```
width: 100%;
```

```
background: #0d1525;
```

```
border: 1px solid var(--border);
```

```
border-radius: 8px;
```

```
padding: 12px 16px;
```

```
font-family: 'Space Mono', monospace;
```

```
font-size: 14px;
```

```
color: var(--text);
```

```
outline: none;
```

```
transition: border-color 0.2s, box-shadow 0.2s; }
```

```
input:focus {
```

```
  border-color: var(--accent);
```

```
  box-shadow: 0 0 0 3px rgba(0,212,255,0.1); }
```

```
input::placeholder { color: var(--muted); }
```

```
.btn {
```

```
  width: 100%;
```

```
  padding: 14px;
```

```
  background: linear-gradient(135deg, var(--accent), #0088aa);
```

```
  border: none;
```

```
  border-radius: 8px;
```

```
  color: #000;
```

```
  font-family: 'Syne', sans-serif;
```

```
  font-size: 15px;
```

```
  font-weight: 700;
```

```
  cursor: pointer;
```

```
  letter-spacing: 0.5px;
```

transition: opacity 0.2s, transform 0.1s;

margin-top: 8px; }

.btn:hover { opacity: 0.9; transform: translateY(-1px); }

.btn:active { transform: translateY(0); }

.btn:disabled { opacity: 0.5; cursor: not-allowed; }

.footer-link { text-align: center; margin-top: 24px; font-size: 14px; color: var(--muted); }

.footer-link a { color: var(--accent); text-decoration: none; font-weight: 600; }

.footer-link a:hover { text-decoration: underline; }

.error-msg {

background: rgba(255,60,110,0.1);

border: 1px solid rgba(255,60,110,0.3);

border-radius: 8px;

padding: 12px 16px;

font-size: 13px;

color: var(--accent2);

margin-bottom: 20px;

```
display: none; }
```

```
.success-msg {
```

```
background: rgba(0,255,136,0.08);
```

```
border: 1px solid rgba(0,255,136,0.25);
```

```
border-radius: 8px;
```

```
padding: 12px 16px;
```

```
font-size: 13px;
```

```
color: var(--green);
```

```
margin-bottom: 20px;
```

```
display: none; }
```

```
</style></head><body>
```

```
<div class="glow-orb orb1"></div>
```

```
<div class="glow-orb orb2"></div>
```

```
<div class="card">
```

```
<div class="logo">
```

```
<div class="logo-icon"><img alt="Shield icon" data-bbox="328 871 348 891"/></div>
```



<div class="logo-text">PhishingGuard</div> </div>

<h2>Welcome back</h2>

<p class="subtitle">Sign in to view your threat dashboard</p>

<div id="error" class="error-msg"></div>

<div id="success" class="success-msg"></div>

<div class="field">

<label>Email Address</label>

<input type="email" id="email" placeholder="you@example.com" autocomplete="email"> </div>

<div class="field">

<label>Password</label>

<input type="password" id="password" placeholder="●●●●●●●●" autocomplete="current-password"></div>

<button class="btn" id="loginBtn" onclick="doLogin()">Sign In →</button>

<div class="footer-link">

Don't have an account? Create one </div> </div>

<script>

```
const API = 'http://localhost:5000';

if (localStorage.getItem('pd_token')) window.location.href = '/dashboard';

async function doLogin() {

  const email = document.getElementById('email').value.trim();

  const password = document.getElementById('password').value;

  const btn = document.getElementById('loginBtn');

  const errEl = document.getElementById('error');

  const sucEl = document.getElementById('success');

  errEl.style.display = 'none';

  sucEl.style.display = 'none';

  if (!email || !password) {

    errEl.textContent = 'Please fill in all fields.';

    errEl.style.display = 'block';

    return;  }

  btn.disabled = true;

  btn.textContent = 'Signing in...';
```

```
try {

  const res = await fetch(`${API}/api/auth/login`, {

    method: 'POST',

    headers: {'Content-Type': 'application/json'},

    body: JSON.stringify({ email, password })  });

  const data = await res.json();

  if (!res.ok) throw new Error(data.error || 'Login failed');

  localStorage.setItem('pd_token', data.token);

  localStorage.setItem('pd_user', JSON.stringify(data.user));

  sucEl.textContent = 'Login successful! Redirecting...';

  sucEl.style.display = 'block';

  setTimeout(() => window.location.href = '/dashboard', 800);

} catch(e) {

  errEl.textContent = e.message;

  errEl.style.display = 'block';

  btn.disabled = false;
```

```
btn.textContent = 'Sign In →';    } }
```

```
document.addEventListener('keypress', e => { if (e.key === 'Enter') doLogin(); });
```

```
</script></body></html>
```

4.Profile.html:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head> <meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Profile — PhishingGuard</title>
```

```
<link rel="preconnect" href="https://fonts.googleapis.com">
```

```
<link  
href="https://fonts.googleapis.com/css2?family=Space+Mono:wght@400;700&family=Syne:wght@  
400;600;700;800&display=swap" rel="stylesheet">
```

```
<style>
```

```
:root { --bg:#0a0e1a; --panel:#111827; --border:#1e2d45; --accent:#00d4ff; --accent2:#ff3c6e; --  
green:#00ff88; --text:#e2e8f0; --muted:#64748b; --sidebar-w:240px; }
```

```
* { margin:0; padding:0; box-sizing:border-box; }
```

```
body { background:var(--bg); font-family:'Syne',sans-serif; color:var(--text); display:flex; min-  
height:100vh; }
```

```
.sidebar { width:var(--sidebar-w); background:var(--panel); border-right:1px solid var(--border);
display:flex; flex-direction:column; position:fixed; height:100vh; z-index:100; }

.sidebar-logo { padding:28px 24px; border-bottom:1px solid var(--border); }

.sidebar-logo .name { font-size:18px; font-weight:800; }

.sidebar-logo .name span { color:var(--accent); }

.sidebar-logo .tagline { font-size:11px; color:var(--muted); margin-top:2px; }

.nav { flex:1; padding:20px 12px; }

.nav-item { display:flex; align-items:center; gap:12px; padding:11px 14px; border-radius:8px; font-
size:14px; font-weight:600; cursor:pointer; color:var(--muted); transition:all .2s; text-
decoration:none; margin-bottom:4px; }

.nav-item:hover { background:rgba(0,212,255,0.07); color:var(--text); }

.nav-item.active { background:rgba(0,212,255,0.12); color:var(--accent); }

.nav-item .icon { font-size:18px; width:22px; text-align:center; }

.sidebar-user { padding:16px 20px; border-top:1px solid var(--border); }

.user-badge { display:flex; align-items:center; gap:10px; }

.user-avatar-sm { width:36px; height:36px; border-radius:50%; background:linear-gradient(135deg,
var(--accent), var(--accent2)); display:flex; align-items:center; justify-content:center; font-
weight:700; font-size:14px; color:#000; flex-shrink:0; }

.user-name { font-size:13px; font-weight:600; }

.user-email { font-size:11px; color:var(--muted); }
```

```
.logout-btn { margin-top:12px; width:100%; padding:8px; background:transparent; border:1px solid var(--border); border-radius:6px; color:var(--muted); font-size:12px; font-family:'Syne',sans-serif; cursor:pointer; transition:all .2s; }
```

```
.logout-btn:hover { border-color:var(--accent2); color:var(--accent2); }
```

```
.main { margin-left:var(--sidebar-w); flex:1; padding:32px; max-width:800px; }
```

```
.page-header { margin-bottom:32px; }
```

```
.page-header h1 { font-size:28px; font-weight:800; }
```

```
.page-header p { color:var(--muted); font-size:14px; margin-top:4px; }
```

```
.profile-card { background:var(--panel); border:1px solid var(--border); border-radius:12px; padding:32px; margin-bottom:24px; }
```

```
.avatar-big { width:80px; height:80px; border-radius:50%; background:linear-gradient(135deg, var(--accent), var(--accent2)); display:flex; align-items:center; justify-content:center; font-weight:800; font-size:32px; color:#000; margin-bottom:16px; }
```

```
.profile-name { font-size:24px; font-weight:800; }
```

```
.profile-email { color:var(--muted); font-size:14px; margin-top:4px; }
```

```
.profile-since { font-size:12px; color:var(--muted); margin-top:8px; font-family:'Space Mono',monospace; }
```

```
.stats-row { display:flex; gap:20px; margin-top:24px; }
```

```
.mini-stat { flex:1; background:#0d1525; border:1px solid var(--border); border-radius:10px; padding:16px; text-align:center; }
```

```
.mini-stat-num { font-size:28px; font-weight:800; font-family:'Space Mono',monospace;
color:var(--accent); }
```

```
.mini-stat-label { font-size:11px; color:var(--muted); margin-top:4px; text-transform:uppercase;
letter-spacing:0.5px; }
```

```
.section-title { font-size:13px; font-weight:700; color:var(--muted); text-transform:uppercase;
letter-spacing:1px; margin-bottom:16px; }
```

```
.field { margin-bottom:18px; }
```

```
label { display:block; font-size:12px; font-weight:600; color:var(--muted); text-
transform:uppercase; letter-spacing:1px; margin-bottom:8px; }
```

```
input {
```

```
width:100%; background:#0d1525; border:1px solid var(--border); border-radius:8px;
```

```
padding:12px 16px; font-family:'Space Mono',monospace; font-size:14px;
```

```
color:var(--text); outline:none; transition:border-color .2s;
```

```
}
```

```
input:focus { border-color:var(--accent); }
```

```
input:disabled { opacity:0.5; cursor:not-allowed; }
```

```
.btn { padding:12px 24px; background:linear-gradient(135deg,var(--accent),#0088aa);
border:none; border-radius:8px; color:#000; font-family:'Syne',sans-serif; font-size:14px; font-
weight:700; cursor:pointer; transition:opacity .2s; }
```

```
.btn:hover { opacity:.85; }
```

```
.btn:disabled { opacity:.5; cursor:not-allowed; }
```

```
.msg { padding:10px 14px; border-radius:6px; font-size:13px; margin-top:12px; display:none; }
```

```
.msg-ok { background:rgba(0,255,136,0.1); border:1px solid rgba(0,255,136,0.25); color:var(--green); }
```

```
.msg-err { background:rgba(255,60,110,0.1); border:1px solid rgba(255,60,110,0.25); color:var(--accent2); }
```

```
</style></head>
```

```
<body>
```

```
<nav class="sidebar">
```

```
<div class="sidebar-logo">
```

```
<div class="name">Phishing<span>Guard</span></div>
```

```
<div class="tagline">Real-time threat protection</div>
```

```
</div>
```

```
<div class="nav">
```

```
<a href="/dashboard" class="nav-item"><span class="icon"> </span> Dashboard</a>
```

```
<a href="/history" class="nav-item"><span class="icon"> </span> Scan History</a>
```

```
<a href="/profile" class="nav-item active"><span class="icon"> </span> Profile</a>
```

```
</div>
```

```
<div class="sidebar-user">
```

```
<div class="user-badge">
```

```
<div class="user-avatar-sm" id="userInitial">?</div>
```

```
<div>
```

```
<div class="user-name" id="sidebarName">Loading...</div>
```

```
<div class="user-email" id="sidebarEmail"></div>
```

```
</div> </div>
```

```
<button class="logout-btn" onclick="logout()">Sign Out</button>
```

```
</div> </nav>
```

```
<main class="main">
```

```
<div class="page-header">
```

```
<h1>My Profile</h1>
```

```
<p>Manage your account details</p> </div>
```

```
<div class="profile-card">
```

```
<div class="avatar-big" id="avatarBig">?</div>
```

```
<div class="profile-name" id="profileName">—</div>
```

```
<div class="profile-email" id="profileEmail">—</div>
```

```
<div class="profile-since" id="profileSince">Member since —</div>
```

```
<div class="stats-row">
```

```
<div class="mini-stat">
```

```
<div class="mini-stat-num" id="pTotal">—</div>
```

```
<div class="mini-stat-label">Total Scans</div>
```

```
</div>
```

```
<div class="mini-stat">
```

```
<div class="mini-stat-num" id="pThreats" style="color:var(--accent2)">—</div>
```

```
<div class="mini-stat-label">Threats Blocked</div>
```

```
</div>
```

```
<div class="mini-stat">
```

```
<div class="mini-stat-num" id="pSafe" style="color:var(--green)">—</div>
```

```
<div class="mini-stat-label">Safe Sites</div>
```

```
</div> </div> </div>
```

```
<div class="profile-card">
```

```
<div class="section-title">Account Information</div>
```

```
<div class="field">
```

```
<label>Full Name</label>
```

```
<input type="text" id="editName" disabled> </div>
```

```
<div class="field">
```

```
<label>Email Address</label>
```

```
<input type="email" id="editEmail" disabled>
```

```
</div>
```

```
<div id="infoMsg" class="msg"></div>
```

```
<div style="display:flex;gap:12px;margin-top:4px">
```

```
<button class="btn" id="editBtn" onclick="toggleEdit()">Edit Profile</button>
```

```
<button class="btn" id="saveBtn" onclick="saveProfile()" style="display:none">Save  
Changes</button> </div> </div> </main>
```

```
<script>
```

```
const API = 'http://localhost:5000';
```

```
const token = localStorage.getItem('pd_token');
```

```
if (!token) window.location.href = '/login';
```

```
function logout() {

    localStorage.removeItem('pd_token'); localStorage.removeItem('pd_user');

    window.location.href = '/login';  }

let editing = false;

function toggleEdit() {

    editing = !editing;

    document.getElementById('editName').disabled = !editing;

    document.getElementById('editEmail').disabled = !editing;

    document.getElementById('editBtn').textContent = editing ? 'Cancel' : 'Edit Profile';

    document.getElementById('saveBtn').style.display = editing ? '' : 'none';  }

async function loadProfile() {

    const res = await fetch(`${API}/api/auth/me`, { headers: {'Authorization': 'Bearer '+token} });

    if (res.status === 401) { logout(); return; }

    const u = await res.json();

    document.getElementById('avatarBig').textContent = u.name[0].toUpperCase();

    document.getElementById('userInitial').textContent = u.name[0].toUpperCase();
```

```
document.getElementById('profileName').textContent = u.name;

document.getElementById('profileEmail').textContent = u.email;

document.getElementById('sidebarName').textContent = u.name;

document.getElementById('sidebarEmail').textContent = u.email;


document.getElementById('profileSince').textContent = 'Member since ' + new
Date(u.created_at).toLocaleDateString();


document.getElementById('editName').value = u.name;

document.getElementById('editEmail').value = u.email;

document.getElementById('pTotal').textContent = u.total_scans;

document.getElementById('pThreats').textContent = u.threats_blocked;

document.getElementById('pSafe').textContent = u.total_scans - u.threats_blocked;  }

function saveProfile() {

    // Placeholder — extend with a PATCH /api/auth/profile endpoint as needed

    const msg = document.getElementById('infoMsg');

    msg.textContent = 'Profile update coming soon!';

    msg.className = 'msg msg-ok';

    msg.style.display = 'block';
```

```
toggleEdit();
```

```
setTimeout(() => msg.style.display = 'none', 3000); }
```

```
loadProfile();
```

```
</script></body></html>
```

5. Resister.html:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Register — Phishing Detector</title>
```

```
<link rel="preconnect" href="https://fonts.googleapis.com">
```

```
<link  
href="https://fonts.googleapis.com/css2?family=Space+Mono:wght@400;700&family=Syne:wght@  
400;600;800&display=swap" rel="stylesheet">
```

```
<style>
```

```
:root {
```

```
--bg: #0a0e1a; --panel: #111827; --border: #1e2d45;

--accent: #00d4ff; --accent2: #ff3c6e; --green: #00ff88;

--text: #e2e8f0; --muted: #64748b;  }

* { margin:0; padding:0; box-sizing:border-box; }

body {

  background: var(--bg); font-family: 'Syne', sans-serif;

  color: var(--text); min-height: 100vh;

  display: flex; align-items: center; justify-content: center; overflow: hidden;  }

body::before {

  content: ""; position: fixed; inset: 0;

  background-image: linear-gradient(rgba(0,212,255,0.04) 1px, transparent 1px), linear-
gradient(90deg, rgba(0,212,255,0.04) 1px, transparent 1px);

  background-size: 40px 40px; pointer-events: none; z-index: 0;  }

.glow-orb { position:fixed; border-radius:50%; filter:blur(80px); pointer-events:none; z-index:0; }

.orb1 { width:400px; height:400px; background:rgba(0,255,136,0.06); top:-100px; right:-80px; }

.orb2 { width:300px; height:300px; background:rgba(0,212,255,0.07); bottom:-80px; left:-80px; }

.card {
```

```
position:relative; z-index:1; background:var(--panel);

border:1px solid var(--border); border-radius:16px;

padding:48px 44px; width:420px;

box-shadow: 0 0 60px rgba(0,0,0,0.5), 0 0 0 1px rgba(0,255,136,0.05);  }

.logo { display:flex; align-items:center; gap:12px; margin-bottom:36px; }

.logo-icon { width:44px; height:44px; background:linear-gradient(135deg, var(--green), var(--
accent)); border-radius:10px; display:flex; align-items:center; justify-content:center; font-size:22px; }

.logo-text { font-size:20px; font-weight:800; letter-spacing:-0.5px; }

.logo-text span { color: var(--green); }

h2 { font-size:26px; font-weight:800; margin-bottom:6px; }

.subtitle { color:var(--muted); font-size:14px; margin-bottom:32px; }

.field { margin-bottom:20px; }

label { display:block; font-size:12px; font-weight:600; color:var(--muted); text-
transform:uppercase; letter-spacing:1px; margin-bottom:8px; }

input {

width:100%; background:#0d1525; border:1px solid var(--border); border-radius:8px;

padding:12px 16px; font-family:'Space Mono',monospace; font-size:14px;
```

```
color:var(--text); outline:none; transition:border-color .2s, box-shadow .2s; }

input:focus { border-color:var(--green); box-shadow:0 0 3px rgba(0,255,136,0.1); }

input::placeholder { color:var(--muted); }

.password-strength { margin-top:8px; height:4px; border-radius:2px; background:var(--border);
overflow:hidden; }

.strength-bar { height:100%; width:0%; transition:width .3s, background .3s; border-radius:2px; }

.btn {

width:100%; padding:14px;

background:linear-gradient(135deg, var(--green), #00aa66);

border:none; border-radius:8px; color:#000;

font-family:'Syne',sans-serif; font-size:15px; font-weight:700;

cursor:pointer; letter-spacing:0.5px; transition:opacity .2s, transform .1s; margin-top:8px; }

.btn:hover { opacity:.9; transform:translateY(-1px); }

.btn:disabled { opacity:.5; cursor:not-allowed; }

.footer-link { text-align:center; margin-top:24px; font-size:14px; color:var(--muted); }

.footer-link a { color:var(--accent); text-decoration:none; font-weight:600; }
```

```
.error-msg { background:rgba(255,60,110,0.1); border:1px solid rgba(255,60,110,0.3); border-radius:8px; padding:12px 16px; font-size:13px; color:var(--accent2); margin-bottom:20px; display:none; }
```

```
.success-msg { background:rgba(0,255,136,0.08); border:1px solid rgba(0,255,136,0.25); border-radius:8px; padding:12px 16px; font-size:13px; color:var(--green); margin-bottom:20px; display:none; }
```

```
</style></head>
```

```
<body>
```

```
<div class="glow-orb orb1"></div>
```

```
<div class="glow-orb orb2"></div>
```

```
<div class="card">
```

```
<div class="logo">
```

```
<div class="logo-icon"><img alt="Phishing Guard logo icon" data-bbox="328 641 348 658"/></div>
```

```
<div class="logo-text">Phishing<span>Guard</span></div> </div>
```

```
<h2>Create account</h2>
```

```
<p class="subtitle">Start tracking and protecting your browsing</p>
```

```
<div id="error" class="error-msg"></div>
```

```
<div id="success" class="success-msg"></div>
```

```
<div class="field">
```

```
<label>Full Name</label>
```

```
<input type="text" id="name" placeholder="John Doe">
```

```
</div>
```

```
<div class="field">
```

```
<label>Email Address</label>
```

```
<input type="email" id="email" placeholder="you@example.com"> </div>
```

```
<div class="field">
```

```
<label>Password</label>
```

```
<input type="password" id="password" placeholder="Minimum 6 characters"
oninput="checkStrength(this.value)">
```

```
<div class="password-strength"><div class="strength-bar"
id="strengthBar"></div></div> </div>
```

```
<button class="btn" id="regBtn" onclick="doRegister()">Create Account →</button>
```

```
<div class="footer-link">
```

```
Already have an account? <a href="/login">Sign in</a>
```

```
</div> </div>
```

```
<script>
```

```
const API = 'http://localhost:5000';
```

```
if (localStorage.getItem('pd_token')) window.location.href = '/dashboard';
```

```
function checkStrength(pw) {
```

```
    const bar = document.getElementById('strengthBar');
```

```
    let score = 0;
```

```
    if (pw.length >= 6) score++;
```

```
    if (pw.length >= 10) score++;
```

```
    if (/[A-Z]/.test(pw)) score++;
```

```
    if (/[0-9]/.test(pw)) score++;
```

```
    if (/^[A-Za-z0-9]/.test(pw)) score++;
```

```
    const pct = (score / 5) * 100;
```

```
    const colors = ['#ff3c6e', '#ff7b3c', '#ffc107', '#00d4ff', '#00ff88'];
```

```
    bar.style.width = pct + '%';
```

```
    bar.style.background = colors[score - 1] || '#ff3c6e'; }
```

```
async function doRegister() {
```

```
const name = document.getElementById('name').value.trim();

const email = document.getElementById('email').value.trim();

const password = document.getElementById('password').value;

const btn = document.getElementById('regBtn');

const errEl = document.getElementById('error');

const sucEl = document.getElementById('success');

errEl.style.display = 'none'; sucEl.style.display = 'none';

if (!name || !email || !password) {

    errEl.textContent = 'Please fill in all fields.'; errEl.style.display = 'block'; return; }

btn.disabled = true; btn.textContent = 'Creating account...';

try {

    const res = await fetch(`${API}/api/auth/register`, {

        method: 'POST',

        headers: {'Content-Type': 'application/json'},

        body: JSON.stringify({ name, email, password })    });

    const data = await res.json();
```

```
if (!res.ok) throw new Error(data.error || 'Registration failed');

localStorage.setItem('pd_token', data.token);

localStorage.setItem('pd_user', JSON.stringify(data.user));

sucEl.textContent = 'Account created! Redirecting to dashboard...';

sucEl.style.display = 'block';

setTimeout(() => window.location.href = '/dashboard', 900);

} catch(e) {

    errEl.textContent = e.message; errEl.style.display = 'block';

    btn.disabled = false; btn.textContent = 'Create Account →';    } }

document.addEventListener('keypress', e => { if (e.key === 'Enter') doRegister(); });

</script></body></html>
```

*****Backend*****

1.App.py:

```
from flask import Flask, request, jsonify, render_template, redirect, url_for

from flask_pymongo import PyMongo

from flask_cors import CORS
```

```
from werkzeug.security import generate_password_hash, check_password_hash
```

```
from datetime import datetime, timedelta
```

```
from bson import ObjectId
```

```
import jwt, os, re
```

```
app = Flask(__name__)
```

```
app.config['SECRET_KEY'] = os.environ.get('SECRET_KEY', 'phishing-secret-2024')
```

```
app.config['MONGO_URI'] = os.environ.get('MONGO_URI',  
'mongodb://localhost:27017/phishing_detector')
```

```
mongo = PyMongo(app)
```

```
CORS(app, resources={r"/api/*": {"origins": "*"}})
```

```
# — JWT helpers —————
```

```
def generate_token(user_id, email):
```

```
    return jwt.encode({
```

```
        'user_id': str(user_id), 'email': email,
```

```
        'exp': datetime.utcnow() + timedelta(hours=24)
```

```
    }, app.config['SECRET_KEY'], algorithm='HS256')
```

```
def verify_token(token):
```

try:

 return jwt.decode(token, app.config['SECRET_KEY'], algorithms=['HS256'])

except:

 return None

def get_current_user():

 h = request.headers.get('Authorization', '')

 if h.startswith('Bearer '):

 p = verify_token(h[7:])

 if p:

 return mongo.db.users.find_one({'_id': ObjectId(p['user_id'])})

 return None

def require_auth(f):

 from functools import wraps

 @wraps(f)

 def decorated(*args, **kwargs):

 user = get_current_user()

if not user:

return jsonify({'error': 'Unauthorized. Please login.'}), 401

return f(user, *args, **kwargs)

return decorated

— Auth routes —————

@app.route('/api/auth/register', methods=['POST'])

def register():

d = request.get_json()

name, email, password = d.get('name', '').strip(), d.get('email', '').strip().lower(), d.get('password', '')

if not name or not email or not password:

return jsonify({'error': 'All fields required'}), 400

if len(password) < 6:

return jsonify({'error': 'Password must be at least 6 characters'}), 400

if not re.match(r'^[^\@]+\@[^\@]+\.[^\@]+\$', email):

return jsonify({'error': 'Invalid email'}), 400

if mongo.db.users.find_one({'email': email}):

```
        return jsonify({'error': 'Email already registered'}), 409

    uid = mongo.db.users.insert_one({

        'name': name, 'email': email,

        'password': generate_password_hash(password),

        'created_at': datetime.utcnow(),

        'total_scans': 0, 'threats_blocked': 0

    }).inserted_id

    token = generate_token(uid, email)

    return jsonify({'message': 'Account created', 'token': token,

                    'user': {'id': str(uid), 'name': name, 'email': email}}), 201

@app.route('/api/auth/login', methods=['POST'])

def login():

    d = request.get_json()

    email, password = d.get('email', '').strip().lower(), d.get('password', '')

    if not email or not password:

        return jsonify({'error': 'Email and password required'}), 400
```

```
user = mongo.db.users.find_one({'email': email})

if not user or not check_password_hash(user['password'], password):

    return jsonify({'error': 'Invalid email or password'}), 401

token = generate_token(user['_id'], email)

return jsonify({'message': 'Login successful', 'token': token,

                'user': {'id': str(user['_id']), 'name': user['name'], 'email': email}})

@app.route('/api/auth/me', methods=['GET'])

@require_auth

def get_me(u):

    return jsonify({'id': str(u['_id']), 'name': u['name'], 'email': u['email'],

                    'created_at': u['created_at'].isoformat(),

                    'total_scans': u.get('total_scans', 0),

                    'threats_blocked': u.get('threats_blocked', 0)})

# — Scan routes —————

@app.route('/api/scans', methods=['POST'])

@require_auth
```

```
def save_scan(u):

    d = request.get_json()

    url, threats = d.get('url',''), d.get('threats',[])

    if not url:

        return jsonify({'error': 'URL required'}), 400

    sevs = [t.get('severity') for t in threats]

    status = 'critical' if 'critical' in sevs else 'high' if 'high' in sevs else 'medium' if 'medium' in sevs else
'low' if threats else 'safe'

    sid = mongo.db.scans.insert_one({

        'user_id': u['_id'], 'url': url, 'threats': threats,

        'threat_count': len(threats), 'status': status,

        'scanned_at': datetime.utcnow()

    }).inserted_id

    mongo.db.users.update_one({'_id': u['_id']}, {'$inc': {

        'total_scans': 1, 'threats_blocked': 1 if threats else 0}})

    return jsonify({'message': 'Scan saved', 'scan_id': str(sid)}), 201

@app.route('/api/scans', methods=['GET'])
```

@require_auth

def get_scans(u):

page, limit = int(request.args.get('page',1)), int(request.args.get('limit',20))

scans = list(mongo.db.scans.find({'user_id': u['_id']}),

sort=[('scanned_at',-1)]).skip((page-1)*limit).limit(limit))

total = mongo.db.scans.count_documents({'user_id': u['_id']})

for s in scans:

s['_id'] = str(s['_id']); s['user_id'] = str(s['user_id'])

s['scanned_at'] = s['scanned_at'].isoformat()

return jsonify({'scans': scans, 'total': total, 'page': page,

'pages': (total + limit - 1) // limit})

@app.route('/api/scans/stats', methods=['GET'])

@require_auth

def get_stats(u):

uid = u['_id']

total = mongo.db.scans.count_documents({'user_id': uid})

```
threats = mongo.db.scans.count_documents({'user_id': uid, 'threat_count': {'$gt': 0}})

bd = {r['_id']: r['count'] for r in mongo.db.scans.aggregate([

    {'$match': {'user_id': uid}},

    {'$group': {'_id': '$status', 'count': {'$sum': 1}}]})

week_ago = datetime.utcnow() - timedelta(days=7)

recent = list(mongo.db.scans.aggregate([

    {'$match': {'user_id': uid, 'scanned_at': {'$gte': week_ago}}},

    {'$group': {'_id': {'$dateToString': {'format': '%Y-%m-%d', 'date': '$scanned_at'}},

        'scans': {'$sum': 1},

        'threats': {'$sum': {'$cond': [{'$gt': ['$threat_count', 0]}, 1, 0]}}},

    {'$sort': {'_id': 1}}]))

return jsonify({'total_scans': total, 'threats_detected': threats,

    'safe_sites': total - threats, 'breakdown': bd, 'recent_activity': recent})

@app.route('/api/scans/<scan_id>', methods=['DELETE'])

@require_auth

def delete_scan(u, scan_id):
```

```
r = mongo.db.scans.delete_one({'_id': ObjectId(scan_id), 'user_id': u['_id']})
```

```
if r.deleted_count == 0:
```

```
    return jsonify({'error': 'Scan not found'}), 404
```

```
return jsonify({'message': 'Deleted'})
```

```
# — Page routes —————
```

```
@app.route('/')
```

```
def index(): return redirect(url_for('dashboard'))
```

```
@app.route('/login')
```

```
def login_page(): return render_template('login.html')
```

```
@app.route('/register')
```

```
def register_page(): return render_template('register.html')
```

```
@app.route('/dashboard')
```

```
def dashboard(): return render_template('dashboard.html')
```

```
@app.route('/history')
```

```
def history(): return render_template('history.html')
```

```
@app.route('/profile')
```

```
def profile(): return render_template('profile.html')
```

```
if __name__ == '__main__':
```

```
    app.run(debug=True, port=5000)
```

*****Extension*****

1.background.js:

```
// Background script runs in the background and monitors all tabs
```

```
// This is the main security checking logic
```

```
// List of suspicious TLDs (Top Level Domains) often used in phishing
```

```
const SUSPICIOUS_TLDS = ['.ru', '.tk', '.xyz', '.top', '.work', '.click', '.gq', '.ml', '.ga', '.cf'];
```

```
// List of commonly phished brands - used to detect lookalike domains
```

```
const POPULAR_BRANDS = [
```

```
    'google', 'facebook', 'amazon', 'paypal', 'microsoft', 'apple',
```

```
    'netflix', 'instagram', 'twitter', 'linkedin', 'ebay', 'yahoo',
```

```
    'bankofamerica', 'chase', 'wellsfargo', 'citibank', 'MNSBANK'
```

```
];
```

```
const SUSPICIOUS_CHARS = {
```

```
'0': 'o', // zero instead of letter o

'1': 'i', // one instead of letter i

'3': 'e', // three instead of letter e

'5': 's', // five instead of letter s

'8': 'b', // eight instead of letter b

'vv': 'w' // double v instead of w};

// Function to check if URL is suspicious

function analyzeURL(url) {

  const threats = [];

  try {

    const urlObj = new URL(url);

    const hostname = urlObj.hostname.toLowerCase();

    const protocol = urlObj.protocol;

    // Check 1: HTTPS validation

    if (protocol !== 'https:') {

      threats.push({
```

```
type: 'NO_HTTPS',
```

```
message: '⚠ Site does not use HTTPS encryption',
```

```
severity: 'medium'   }); }
```

```
// Check 2: Suspicious TLD
```

```
for (const tld of SUSPICIOUS_TLDS) {
```

```
  if (hostname.endsWith(tld)) {
```

```
    threats.push({
```

```
      type: 'SUSPICIOUS_TLD',
```

```
      message: `🚩 Suspicious domain extension: ${tld}`,
```

```
      severity: 'high'   });
```

```
    break;  } }
```

```
// Check 3: Excessive length (phishing URLs are often very long)
```

```
if (url.length > 100) {
```

```
  threats.push({
```

```
    type: 'LONG_URL',
```

```
    message: '⚠ Unusually long URL detected',
```

```
severity: 'medium'    }); }
```

```
// Check 4: Too many subdomains (e.g., secure.login.paypal.suspicious.com)
```

```
const subdomains = hostname.split('.');
```

```
if (subdomains.length > 4) {
```

```
  threats.push({
```

```
    type: 'MANY_SUBDOMAINS',
```

```
    message: '⚠ Excessive subdomains detected',
```

```
    severity: 'medium'    }); }
```

```
// Check 5: IP address instead of domain name
```

```
const ipPattern = /^(\d{1,3}\.){3}\d{1,3}$/;
```

```
if (ipPattern.test(hostname)) {
```

```
  threats.push({
```

```
    type: 'IP_ADDRESS',
```

```
    message: '🚫 Using IP address instead of domain name',
```

```
    severity: 'high'    }); }
```

```
for (const brand of POPULAR_BRANDS) {
```

```
if (hostname.includes(brand)) {

    // Check for character substitutions

    let hasSubstitution = false;

    for (const [fake, real] of Object.entries(SUSPICIOUS_CHARS)) {

        if (hostname.includes(fake) && hostname.includes(brand)) {

            hasSubstitution = true;

            threats.push({

                type: 'LOOKALIKE_DOMAIN',

                message: `🚩 Possible fake ${brand} site (character substitution detected)`,

                severity: 'critical'    });

            break;    }    }

    // Check for brand name in subdomain (e.g., paypal.fake-site.com)

    const parts = hostname.split('.');

    const mainDomain = parts.length >= 2 ? parts[parts.length - 2] : '';

    if (hostname.includes(brand) && !mainDomain.includes(brand)) {

        threats.push({
```

```
type: 'BRAND_IN_SUBDOMAIN',
```

```
message: `⚠ Brand name "${brand}" appears in subdomain, not main domain`,
```

```
severity: 'high'
```

```
});    }  }
```

```
// Check 7: Suspicious keywords in URL
```

```
const suspiciousKeywords = ['verify', 'account', 'secure', 'login', 'update', 'confirm', 'banking'];
```

```
const urlLower = url.toLowerCase();
```

```
for (const keyword of suspiciousKeywords) {
```

```
  if (urlLower.includes(keyword) && urlLower.includes('-')) {
```

```
    threats.push({
```

```
      type: 'SUSPICIOUS_KEYWORDS',
```

```
      message: `⚠ Suspicious keywords with hyphens detected: "${keyword}"`,
```

```
      severity: 'medium'
```

```
    });
```

```
    break;  } }
```

```
// Check 8: Too many hyphens (common in phishing)
```

```
const hyphenCount = (hostname.match(/-/g) || []).length;

if (hyphenCount > 3) {

  threats.push({

    type: 'MANY_HYPHENS',

    message: '⚠ Excessive hyphens in domain name',

    severity: 'medium'

  });

} catch (error) {

  console.error('Error analyzing URL:', error);

}

return threats;

}

// Listen for tab updates (when user navigates to a new page)

chrome.tabs.onUpdated.addListener((tabId, changeInfo, tab) => {

  // Only check when page has finished loading

  if (changeInfo.status === 'complete' && tab.url) {

    const threats = analyzeURL(tab.url);
```

```
// If threats detected, show warning

if (threats.length > 0) {

    // Store threats for popup to access

    chrome.storage.local.set({

        [`threats_${tabId}`]: threats,

        [`url_${tabId}`]: tab.url    });

    // Send message to content script to show warning banner

    chrome.tabs.sendMessage(tabId, {

        action: 'showWarning',

        threats: threats,

        url: tab.url

    }).catch(err => {

        // Ignore errors if content script not ready

        console.log('Could not send message to content script');

    });

    // Update extension icon to show warning
```

```
chrome.action.setBadgeText({ text: '!', tabId: tabId });

chrome.action.setBadgeBackgroundColor({ color: '#FF0000', tabId: tabId });

    // Show browser notification for critical threats

const criticalThreats = threats.filter(t => t.severity === 'critical');

if (criticalThreats.length > 0) {

    chrome.notifications.create({

        type: 'basic',

        imageUrl: 'icon.png',

        title: '⚠️ PHISHING WARNING',

        message: `This site may be dangerous!\n${criticalThreats[0].message}`,

        priority: 2

    });    }

} else {

    // Clear badge if no threats

    chrome.action.setBadgeText({ text: '', tabId: tabId });

    chrome.storage.local.remove(['threats_${tabId}', `url_${tabId}`]);
```

```
    } }));
```

```
// Listen for messages from popup
```

```
chrome.runtime.onMessage.addListener((request, sender, sendResponse) => {
```

```
    if (request.action === 'analyzeCurrentTab') {
```

```
        chrome.tabs.query({ active: true, currentWindow: true }, (tabs) => {
```

```
            if (tabs[0]) {
```

```
                const threats = analyzeURL(tabs[0].url);
```

```
                sendResponse({ threats: threats, url: tabs[0].url });
```

```
            } }));
```

```
    return true; // Keep message channel open for async response } }));
```

2.content.js:

```
// Content script - runs on every webpage
```

```
// This creates the warning banner that appears at the top of suspicious sites
```

```
// Listen for messages from background script
```

```
chrome.runtime.onMessage.addListener((request, sender, sendResponse) => {
```

```
if (request.action === 'showWarning') {

    showWarningBanner(request.threats, request.url); } });

// Function to create and display warning banner

function showWarningBanner(threats, url) {

    // Remove existing banner if present

    const existing = document.getElementById('phishing-warning-banner');

    if (existing) {

        existing.remove(); }

    // Determine overall severity level

    const hasCritical = threats.some(t => t.severity === 'critical');

    const hasHigh = threats.some(t => t.severity === 'high');

    let severityLevel = 'medium';

    let bannerColor = '#FFA500'; // Orange for medium

    let warningText = '⚠️ WARNING: This site may be suspicious';

    if (hasCritical) {

        severityLevel = 'critical';
```

```
bannerColor = '#DC143C'; // Red for critical
```

```
warningText = '🚨 DANGER: This site is likely a phishing attempt!';
```

```
} else if (hasHigh) {
```

```
severityLevel = 'high';
```

```
bannerColor = '#FF6347'; // Tomato red for high
```

```
warningText = '⚠️ ALERT: This site shows multiple warning signs'; }
```

```
// Create banner element
```

```
const banner = document.createElement('div');
```

```
banner.id = 'phishing-warning-banner';
```

```
banner.style.cssText = `
```

```
position: fixed;
```

```
top: 0;
```

```
left: 0;
```

```
width: 100%;
```

```
background: ${bannerColor};
```

```
color: white;
```

padding: 15px 20px;

font-family: Arial, sans-serif;

font-size: 14px;

z-index: 999999;

box-shadow: 0 2px 10px rgba(0,0,0,0.3);

border-bottom: 3px solid rgba(0,0,0,0.2); `;

// Create banner content

const content = document.createElement('div');

content.style.cssText = 'max-width: 1200px; margin: 0 auto; display: flex; justify-content: space-between; align-items: center;';

// Warning message and details

const messageDiv = document.createElement('div');

messageDiv.style.flex = '1';

const title = document.createElement('div');

title.style.cssText = 'font-weight: bold; font-size: 16px; margin-bottom: 8px;';

title.textContent = warningText;

const detailsList = document.createElement('ul');

```
detailsList.style.cssText = 'margin: 5px 0 0 20px; padding: 0; font-size: 13px;';
```

```
// Show top 3 threats
```

```
threats.slice(0, 3).forEach(threat => {
```

```
    const li = document.createElement('li');
```

```
    li.textContent = threat.message;
```

```
    li.style.marginBottom = '3px';
```

```
    detailsList.appendChild(li); });
```

```
messageDiv.appendChild(title);
```

```
messageDiv.appendChild(detailsList);
```

```
// Close button
```

```
const closeBtn = document.createElement('button');
```

```
closeBtn.textContent = '× Close';
```

```
closeBtn.style.cssText = `
```

```
    background: rgba(255,255,255,0.2);
```

```
    border: 2px solid white;
```

```
    color: white;
```

padding: 8px 15px;

border-radius: 5px;

cursor: pointer;

font-size: 14px;

font-weight: bold;

margin-left: 20px; `;

closeBtn.onmouseover = () => closeBtn.style.background = 'rgba(255,255,255,0.3)';

closeBtn.onmouseout = () => closeBtn.style.background = 'rgba(255,255,255,0.2)';

closeBtn.onclick = () => banner.remove();

content.appendChild(messageDiv);

content.appendChild(closeBtn);

banner.appendChild(content);

document.body.insertBefore(banner, document.body.firstChild);

// Add a small delay to push page content down smoothly

document.body.style.marginTop = '0px';

setTimeout(() => {

```
document.body.style.transition = 'margin-top 0.3s ease';

document.body.style.marginTop = banner.offsetHeight + 'px';

}, 10);}

// (in case background script already detected threat)

window.addEventListener('load', () => {

  chrome.runtime.sendMessage({ action: 'checkThreats' });

});
```

3.manifest.json:

```
{

  "manifest_version": 3,

  "name": "Phishing Detection Extension",

  "version": "1.0",

  "description": "Detects and warns users about potentially phishing websites",

  "permissions": [

    "activeTab",

    "tabs",
```

```
"notifications",

"scripting" ],

"host_permissions": [

"<all_urls>" ],

"background": {

"service_worker": "background.js" },

"action": {

"default_popup": "popup.html",

"default_icon": {

"16": "icon.png",

"48": "icon.png",

"128": "icon.png"  } },

"content_scripts": [ {

"matches": [<all_urls>],

"js": ["content.js"],

"run_at": "document_start"  } ],
```

```
"icons": {  
  
  "16": "icon.png",  
  
  "48": "icon.png",  
  
  "128": "icon.png" } }
```

4.popup.html:

```
<!DOCTYPE html>  
  
<html>  
  
  <head>  
  
    <meta charset="UTF-8">  
  
    <title>PhishingGuard</title>  
  
    <link rel="stylesheet" href="styles.css">  
  
  </head>  
  
  <body>  
  
    <div class="container">  
  
      <div class="header">  
  
        <div class="header-icon"></div>
```

```
<h1>PhishingGuard</h1> </div>
```

```
<!-- AUTH SCREEN -->
```

```
<div id="auth-screen" style="display:none;">
```

```
<div class="auth-tabs">
```

```
<button class="auth-tab active" id="tab-login">Login</button>
```

```
<button class="auth-tab" id="tab-register">Register</button> </div>
```

```
<div id="login-form" class="auth-form">
```

```
<div class="field-group">
```

```
<label>Email</label>
```

```
<input type="email" id="login-email" placeholder="you@example.com"> </div>
```

```
<div class="field-group">
```

```
<label>Password</label>
```

```
<input type="password" id="login-password" placeholder="••••••••"> </div>
```

```
<div id="login-error" class="auth-error"></div>
```

```
<button class="auth-btn" id="login-btn">Sign In →</button> </div>
```

```
<div id="register-form" class="auth-form" style="display:none;">
```

```
<div class="field-group">
```

```
<label>Full Name</label>
```

```
<input type="text" id="reg-name" placeholder="John Doe"> </div>
```

```
<div class="field-group">
```

```
<label>Email</label>
```

```
<input type="email" id="reg-email" placeholder="you@example.com"> </div>
```

```
<div class="field-group">
```

```
<label>Password</label>
```

```
<input type="password" id="reg-password" placeholder="Min. 6 characters"> </div>
```

```
<div id="reg-error" class="auth-error"></div>
```

```
<button class="auth-btn" id="reg-btn">Create Account →</button> </div> </div>
```

```
<!-- MAIN SCREEN -->
```

```
<div id="main-screen" style="display:none;">
```

```
<div class="user-bar">
```

```
<span class="user-greeting">👤 <span id="user-name">User</span></span>
```

```
<div style="display:flex;gap:6px;">
```

```
<button class="icon-btn" id="dashboard-btn" title="Open Dashboard"> </button>
```

```
<button class="icon-btn" id="logout-btn" title="Sign Out"> </button>    </div>    </div>
```

```
<div id="status-container">
```

```
<div id="status" class="status-safe">
```

```
<div class="status-icon"> </div>
```

```
<div class="status-text">Analyzing...</div>
```

```
</div>    </div>
```

```
<div class="url-section">
```

```
<div class="label">Current URL:</div>
```

```
<div id="current-url" class="url-text">Loading...</div>
```

```
</div>
```

```
<div id="threats-container" style="display:none;">
```

```
<div class="section-title"> Detected Issues:</div>
```

```
<div id="threats-list"></div>    </div>
```

```
<div id="safe-message" style="display:none;">
```

```
<div class="safe-box">
```

<div class="safe-icon">✓</div>

<p>No phishing indicators detected.</p>

<p class="safe-note">Always verify URLs before entering sensitive info.</p> </div> </div>

<div class="info-section">

<details>

<summary>What do we check?</summary>

<ul class="info-list">

✓ HTTPS encryption

✓ Suspicious domain extensions

✓ Lookalike domains

✓ IP addresses instead of domain names

✓ Suspicious keywords and patterns

 </details> </div> </div>

<div class="footer">

<small>💡 Tip: Always double-check URLs before entering passwords!</small> </div> </div>

```
<script src="popup.js"></script>
```

```
</body> </html>
```

5.popup.js:

```
const API_BASE = 'http://localhost:5000';
```

```
document.addEventListener('DOMContentLoaded', () => {
```

```
    document.getElementById('tab-login').addEventListener('click', () => showTab('login'));
```

```
    document.getElementById('tab-register').addEventListener('click', () => showTab('register'));
```

```
    document.getElementById('login-btn').addEventListener('click', doLogin);
```

```
    document.getElementById('reg-btn').addEventListener('click', doRegister);
```

```
    document.getElementById('dashboard-btn').addEventListener('click', openDashboard);
```

```
    document.getElementById('logout-btn').addEventListener('click', doLogout);
```

```
    document.getElementById('login-password').addEventListener('keypress', e => {
```

```
        if (e.key === 'Enter') doLogin();
```

```
    });
```

```
    document.getElementById('reg-password').addEventListener('keypress', e => {
```

```
        if (e.key === 'Enter') doRegister(); });
```

```
// Check auth state

const token = localStorage.getItem('pd_token');

if (token) {

  showMainScreen();

  analyzeCurrentPage();

} else {

  showAuthScreen(); } });

function showAuthScreen() {

  document.getElementById('auth-screen').style.display = 'block';

  document.getElementById('main-screen').style.display = 'none'; }

function showMainScreen() {

  document.getElementById('auth-screen').style.display = 'none';

  document.getElementById('main-screen').style.display = 'block';

  const user = JSON.parse(localStorage.getItem('pd_user') || '{}');

  document.getElementById('user-name').textContent = user.name || 'User'; }

function showTab(tab) {
```

```
document.getElementById('login-form').style.display = tab === 'login' ? 'block' : 'none';
```

```
document.getElementById('register-form').style.display = tab === 'register' ? 'block' : 'none';
```

```
document.getElementById('tab-login').classList.toggle('active', tab === 'login');
```

```
document.getElementById('tab-register').classList.toggle('active', tab === 'register'); }
```

```
function openDashboard() {
```

```
    chrome.tabs.create({ url: `${API_BASE}/dashboard` }); }
```

```
function doLogout() {
```

```
    localStorage.removeItem('pd_token');
```

```
    localStorage.removeItem('pd_user');
```

```
    showAuthScreen(); }
```

```
// — Login —————
```

```
async function doLogin() {
```

```
    const email = document.getElementById('login-email').value.trim();
```

```
    const password = document.getElementById('login-password').value;
```

```
    const errEl = document.getElementById('login-error');
```

```
    const btn = document.getElementById('login-btn');
```

```
errEl.textContent = '';

if (!email || !password) { errEl.textContent = 'Please fill in all fields.'; return; }

btn.disabled = true;

btn.textContent = 'Signing in...';

try {

  const res = await fetch(`${API_BASE}/api/auth/login`, {

    method: 'POST',

    headers: { 'Content-Type': 'application/json' },

    body: JSON.stringify({ email, password })  });

  const data = await res.json();

  if (!res.ok) throw new Error(data.error || 'Login failed');

  localStorage.setItem('pd_token', data.token);

  localStorage.setItem('pd_user', JSON.stringify(data.user));

  showMainScreen();

  analyzeCurrentPage();

} catch (e) {
```

```
errEl.textContent = e.message;

btn.disabled = false;

btn.textContent = 'Sign In →';

async function doRegister() {

  const name  = document.getElementById('reg-name').value.trim();

  const email  = document.getElementById('reg-email').value.trim();

  const password = document.getElementById('reg-password').value;

  const errEl  = document.getElementById('reg-error');

  const btn    = document.getElementById('reg-btn');

  errEl.textContent = "";

  if (!name || !email || !password) { errEl.textContent = 'Please fill in all fields.'; return; }

  btn.disabled = true;

  btn.textContent = 'Creating account...';

  try {

    const res = await fetch(`${API_BASE}/api/auth/register`, {

      method: 'POST',
```

```
headers: { 'Content-Type': 'application/json' },

body: JSON.stringify({ name, email, password })  });

const data = await res.json();

if (!res.ok) throw new Error(data.error || 'Registration failed');

localStorage.setItem('pd_token', data.token);

localStorage.setItem('pd_user', JSON.stringify(data.user));

showMainScreen();

analyzeCurrentPage();

} catch (e) {

  errEl.textContent = e.message;

  btn.disabled = false;

  btn.textContent = 'Create Account →'; }  }

// — Scan —————

function analyzeCurrentPage() {

  chrome.tabs.query({ active: true, currentWindow: true }, (tabs) => {

    if (!tabs[0]) return;
```

```
const url = tabs[0].url;

displayURL(url);

if (url.startsWith('chrome://') || url.startsWith('edge://') || url.startsWith('about:')) {

    showSpecialPageMessage();

    return; }

chrome.runtime.sendMessage({ action: 'analyzeCurrentTab' }, (response) => {

    if (chrome.runtime.lastError) {

        console.log('Runtime error:', chrome.runtime.lastError.message);

        return; }

    if (response && response.threats !== undefined) {

        displayResults(response.threats);

        saveScanToBackend(url, response.threats);

    } }); });

async function saveScanToBackend(url, threats) {

    const token = localStorage.getItem('pd_token');

    if (!token) return;
```

```
try {  
  
  await fetch(`${API_BASE}/api/scans`, {  
  
    method: 'POST',  
  
    headers: {  
  
      'Content-Type': 'application/json',  
  
      'Authorization': 'Bearer ' + token    },  
  
    body: JSON.stringify({ url, threats })  
  
  });  
  
} catch (e) {  
  
  console.log('Could not save scan:', e.message);  
  
}    }
```

```
function displayURL(url) {  
  
  const el = document.getElementById('current-url');  
  
  try {  
  
    el.textContent = new URL(url).hostname;  
  
    el.title = url;
```

```
} catch {
```

```
    el.textContent = url; } }
```

```
function displayResults(threats) {
```

```
    const statusDiv    = document.getElementById('status');
```

```
    const statusText   = statusDiv.querySelector('.status-text');
```

```
    const statusIcon   = statusDiv.querySelector('.status-icon');
```

```
    const threatsContainer = document.getElementById('threats-container');
```

```
    const threatsList    = document.getElementById('threats-list');
```

```
    const safeMessage    = document.getElementById('safe-message');
```

```
    if (threats.length === 0) {
```

```
        statusDiv.className      = 'status-safe';
```

```
        statusIcon.textContent    = '✓';
```

```
        statusText.textContent    = 'Site appears safe';
```

```
        safeMessage.style.display = 'block';
```

```
        threatsContainer.style.display = 'none';
```

```
        return;    }
```

```
const hasCritical = threats.some(t => t.severity === 'critical');

const hasHigh    = threats.some(t => t.severity === 'high');

if (hasCritical) {

    statusDiv.className = 'status-danger';

    statusIcon.textContent = '⚠';

    statusText.textContent = 'DANGER: Likely phishing site!';

} else if (hasHigh) {

    statusDiv.className = 'status-warning';

    statusIcon.textContent = '⚠';

    statusText.textContent = 'Warning: Suspicious site detected';

} else {

    statusDiv.className = 'status-caution';

    statusIcon.textContent = '!';

    statusText.textContent = 'Caution: Minor concerns detected';    }

safeMessage.style.display = 'none';

threatsContainer.style.display = 'block';
```

```
threatsList.innerHTML = '';
```

```
threats.forEach(threat => {
```

```
    const item = document.createElement('div');
```

```
    item.className = `threat-item threat-${threat.severity}`;
```

```
    const badge = document.createElement('span');
```

```
    badge.className = 'severity-badge';
```

```
    badge.textContent = threat.severity.toUpperCase();
```

```
    const msg = document.createElement('span');
```

```
    msg.className = 'threat-message';
```

```
    msg.textContent = threat.message;
```

```
    item.appendChild(badge);
```

```
    item.appendChild(msg);
```

```
    threatsList.appendChild(item);
```

```
}); }
```

```
function showSpecialPageMessage() {
```

```
    const statusDiv = document.getElementById('status');
```

```
statusDiv.className = 'status-info';
```

```
statusDiv.querySelector('.status-icon').textContent = 'i';
```

```
statusDiv.querySelector('.status-text').textContent = 'Cannot analyze browser pages';
```

```
const safeMessage = document.getElementById('safe-message');
```

```
safeMessage.style.display = 'block';
```

```
safeMessage.innerHTML = `
```

```
<div class="safe-box">
```

```
<div class="safe-icon">❌</div>
```

```
<p>This is a browser internal page.</p>
```

```
<p class="safe-note">Extension works on regular websites only.</p> </div>`;
```

```
document.getElementById('threats-container').style.display = 'none'; }
```

6.style.css:

```
* { margin:0; padding:0; box-sizing:border-box; }
```

```
body {
```

```
font-family: -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, Arial, sans-serif;
```

```
width: 350px;
```

```
background: #f5f5f5;
```

```
color: #333; }
```

```
.container { background: white; min-height: 400px; }
```

```
.header {
```

```
background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
```

```
color: white; padding: 20px; text-align: center;
```

```
box-shadow: 0 2px 10px rgba(0,0,0,0.1); }
```

```
.header-icon { font-size: 40px; margin-bottom: 10px; }
```

```
.header h1 { font-size: 20px; font-weight: 600; margin: 0; }
```

```
.auth-tabs { display: flex; border-bottom: 2px solid #e9ecef; }
```

```
.auth-tab {
```

```
flex: 1; padding: 14px; background: white; border: none;
```

```
font-size: 14px; font-weight: 600; cursor: pointer; color: #6c757d;
```

```
transition: all .2s; border-bottom: 3px solid transparent; margin-bottom: -2px; }
```

```
.auth-tab.active { color: #667eea; border-bottom-color: #667eea; }
```

```
.auth-tab:hover { background: #f8f9fa; }
```

```
.auth-form { padding: 20px; }
```

```
.field-group { margin-bottom: 14px; }
```

```
.field-group label {
```

```
display: block; font-size: 11px; font-weight: 600;
```

```
color: #6c757d; text-transform: uppercase; letter-spacing: 0.5px; margin-bottom: 6px; }
```

```
.field-group input {
```

```
width: 100%; padding: 10px 12px; border: 1.5px solid #dee2e6;
```

```
border-radius: 7px; font-size: 13px; outline: none; transition: border-color .2s; }
```

```
.field-group input:focus { border-color: #667eea; box-shadow: 0 0 3px rgba(102,126,234,0.1); }
```

```
.auth-btn {
```

```
width: 100%; padding: 12px; margin-top: 6px;
```

```
background: linear-gradient(135deg, #667eea, #764ba2);
```

```
border: none; border-radius: 8px; color: white;
```

```
font-size: 14px; font-weight: 700; cursor: pointer; transition: opacity .2s; }
```

```
.auth-btn:hover { opacity: .9; }
```

```
.auth-btn:disabled { opacity: .5; cursor: not-allowed; }
```

```
.auth-error {

    font-size: 12px; color: #dc3545; background: #fff0f0;

    border: 1px solid #f5c6cb; border-radius: 6px;

    padding: 8px 10px; margin-bottom: 10px;    }

.auth-error:empty { display: none; }

.user-bar {

    display: flex; justify-content: space-between; align-items: center;

    padding: 10px 16px; background: #f8f9fa; border-bottom: 1px solid #e9ecef;    }

.user-greeting { font-size: 13px; font-weight: 600; color: #495057; }

.icon-btn {

    background: none; border: 1px solid #dee2e6; border-radius: 6px;

    padding: 4px 8px; cursor: pointer; font-size: 14px; transition: all .2s;    }

.icon-btn:hover { background: #e9ecef; }

/* Status */

#status-container { padding: 20px; background: white; }

#status {
```

```
display: flex; align-items: center; padding: 15px; border-radius: 8px;

font-weight: 600; font-size: 16px; transition: all .3s;

}

.status-icon {

font-size: 24px; margin-right: 12px; width: 30px; height: 30px;

display: flex; align-items: center; justify-content: center;

border-radius: 50%; background: rgba(255,255,255,0.2);

}

.status-safe { background:#d4edda; color:#155724; border:2px solid #c3e6cb; }

.status-caution { background:#fff3cd; color:#856404; border:2px solid #ffeeba; }

.status-warning { background:#ffe4cc; color:#d9480f; border:2px solid #ffc299; }

.status-danger { background:#f8d7da; color:#721c24; border:2px solid #f5c6cb; animation:pulse 2s
infinite; }

.status-info { background:#d1ecf1; color:#0c5460; border:2px solid #bee5eb; }

@keyframes pulse {

0%, 100% { transform:scale(1); }

50% { transform:scale(1.02); } }
```

```
/* URL section */
```

```
.url-section {
```

```
padding: 15px 20px; background: #f8f9fa;
```

```
border-top: 1px solid #e9ecef; border-bottom: 1px solid #e9ecef; }
```

```
.label { font-size:11px; text-transform:uppercase; color:#6c757d; font-weight:600; margin-bottom:5px; }
```

```
.url-text { font-size:14px; color:#495057; word-break:break-all; font-family:'Courier New',monospace; }
```

```
#threats-container { padding: 20px; }
```

```
.section-title { font-size:14px; font-weight:600; color:#495057; margin-bottom:12px; text-transform:uppercase; }
```

```
#threats-list { display:flex; flex-direction:column; gap:10px; }
```

```
.threat-item { display:flex; align-items:flex-start; padding:12px; border-radius:6px; font-size:13px; border-left:4px solid; background:#f8f9fa; }
```

```
.threat-critical { border-left-color:#dc3545; background:#ffe6e6; }
```

```
.threat-high { border-left-color:#ff6347; background:#fff0ed; }
```

```
.threat-medium { border-left-color:#ffc107; background:#fff9e6; }
```

```
.severity-badge {
```

```
display:inline-block; padding:2px 8px; border-radius:3px;
```

```
font-size:10px; font-weight:700; margin-right:10px; min-width:60px; text-align:center;    }

.threat-critical .severity-badge { background:#dc3545; color:white; }

.threat-high    .severity-badge { background:#ff6347; color:white; }

.threat-medium  .severity-badge { background:#ffc107; color:#333; }

.threat-message { flex:1; line-height:1.5; color:#495057

#safe-message { padding: 20px; }

.safe-box { text-align:center; padding:20px; background:#d4edda; border-radius:8px; border:2px
solid #c3e6cb; }

.safe-icon { font-size:48px; margin-bottom:10px; }

.safe-box p { margin:8px 0; color:#155724; font-size:14px; }

.safe-note { font-size:12px !important; color:#5a9c5f !important; font-style:italic; }

/* Info section */

.info-section { padding:15px 20px; background:#f8f9fa; border-top:1px solid #e9ecef; }

details { cursor:pointer; }

summary { font-size:13px; font-weight:600; color:#667eea; padding:8px 0; list-style:none;
display:flex; align-items:center; }

summary::before { content:'►'; margin-right:8px; font-size:10px; transition:transform .2s; }
```

```
details[open] summary::before { transform:rotate(90deg); }
```

```
.info-list { margin-top:12px; padding-left:20px; font-size:12px; line-height:1.8; color:#6c757d; }
```

```
/* Footer */
```

```
.footer { padding:15px 20px; background:#667eea; color:white; text-align:center; }
```

```
.footer small { font-size:12px; opacity:.95; }/*
```

```
##  STEP 3 — Backend Files (ALL NEW, create from scratch
```

```
### FILE 4 — `backend/requirements.txt` (NEW FILE)*/
```

```
flask == 3.0.0
```

```
flask-pymongo==2.3.0
```

```
flask-cors==4.0.0
```

```
werkzeug==3.0.1
```

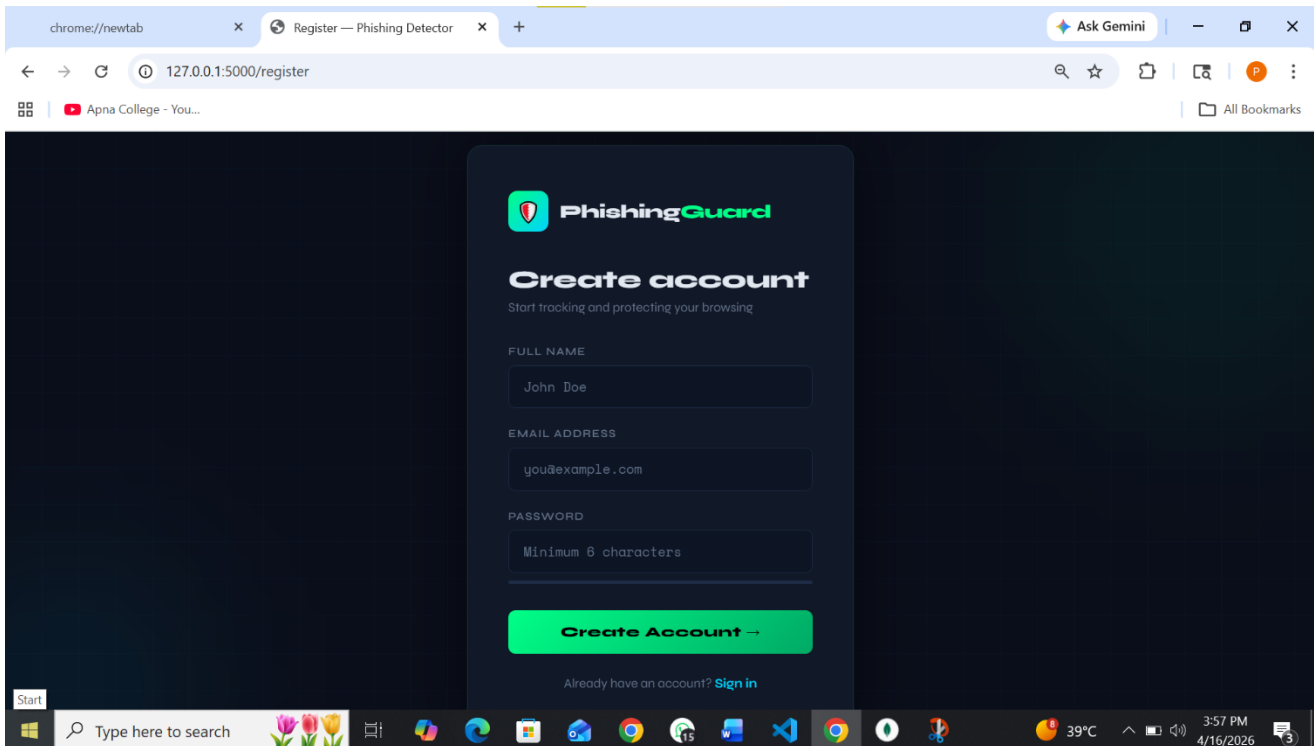
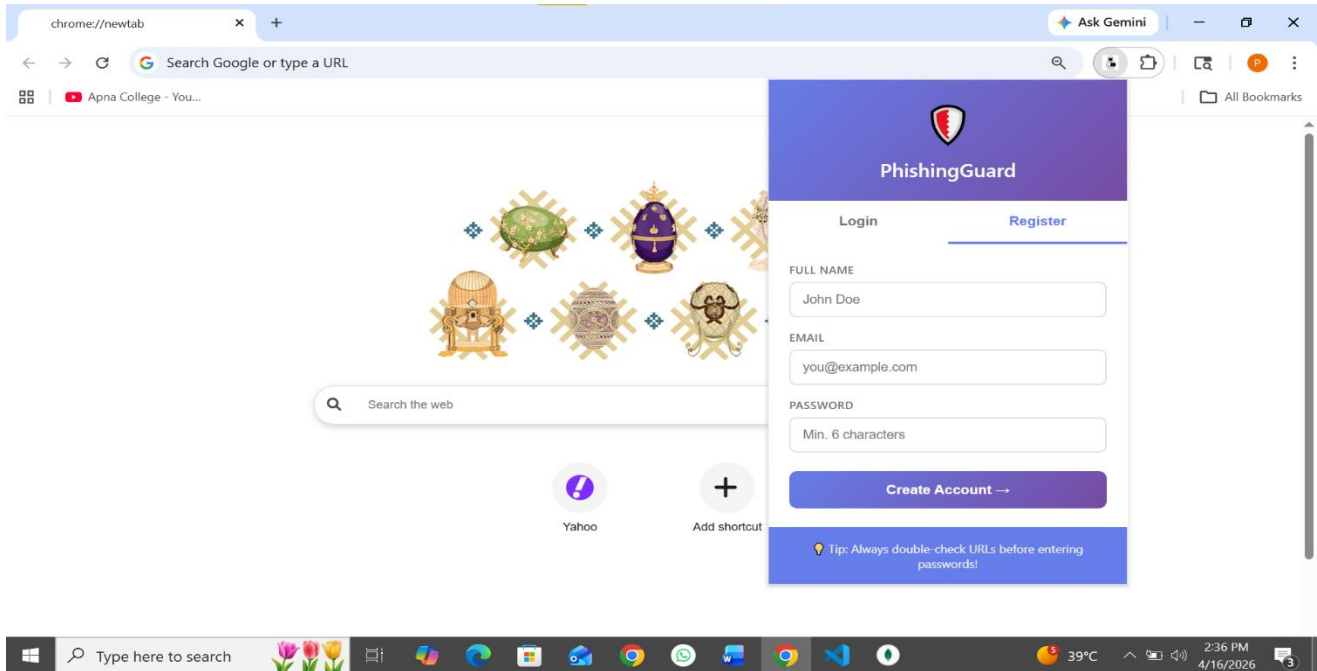
```
PyJWT==2.8.0
```

```
pymongo==4.6.1
```

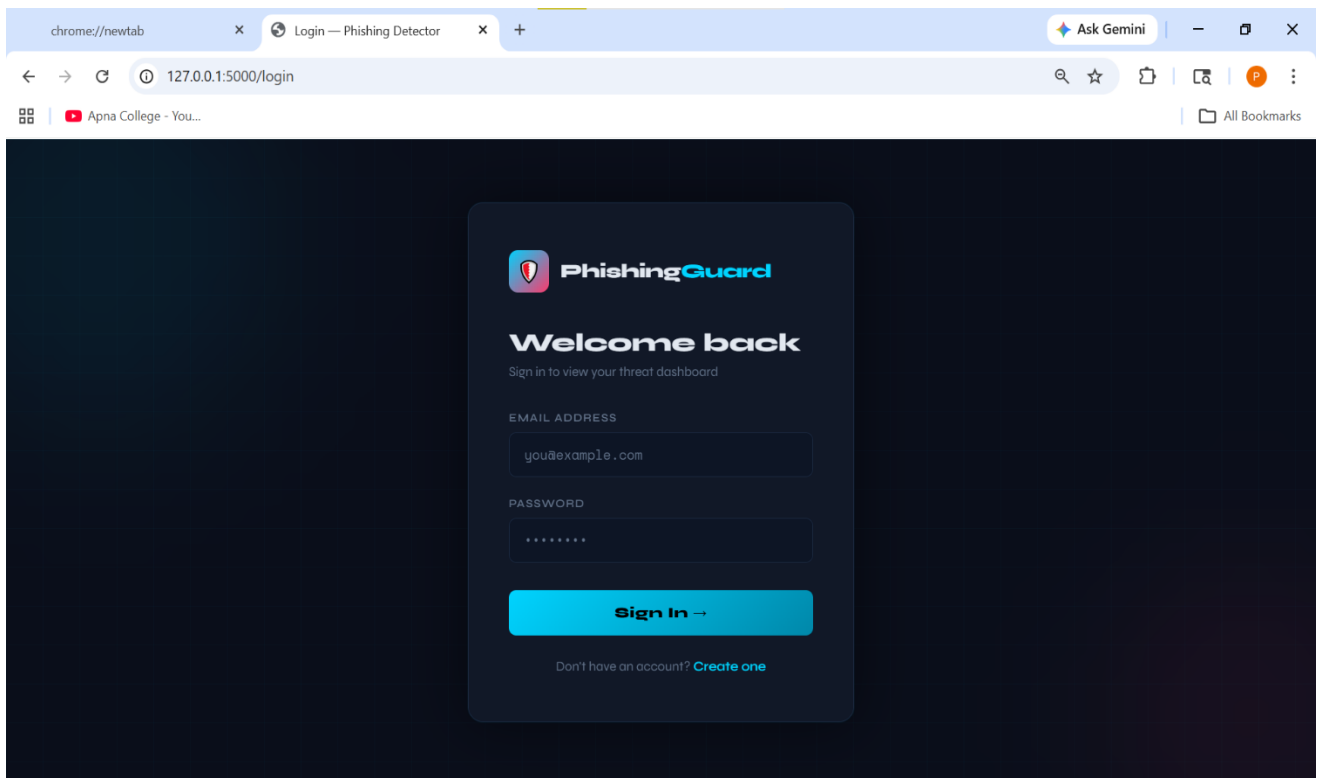
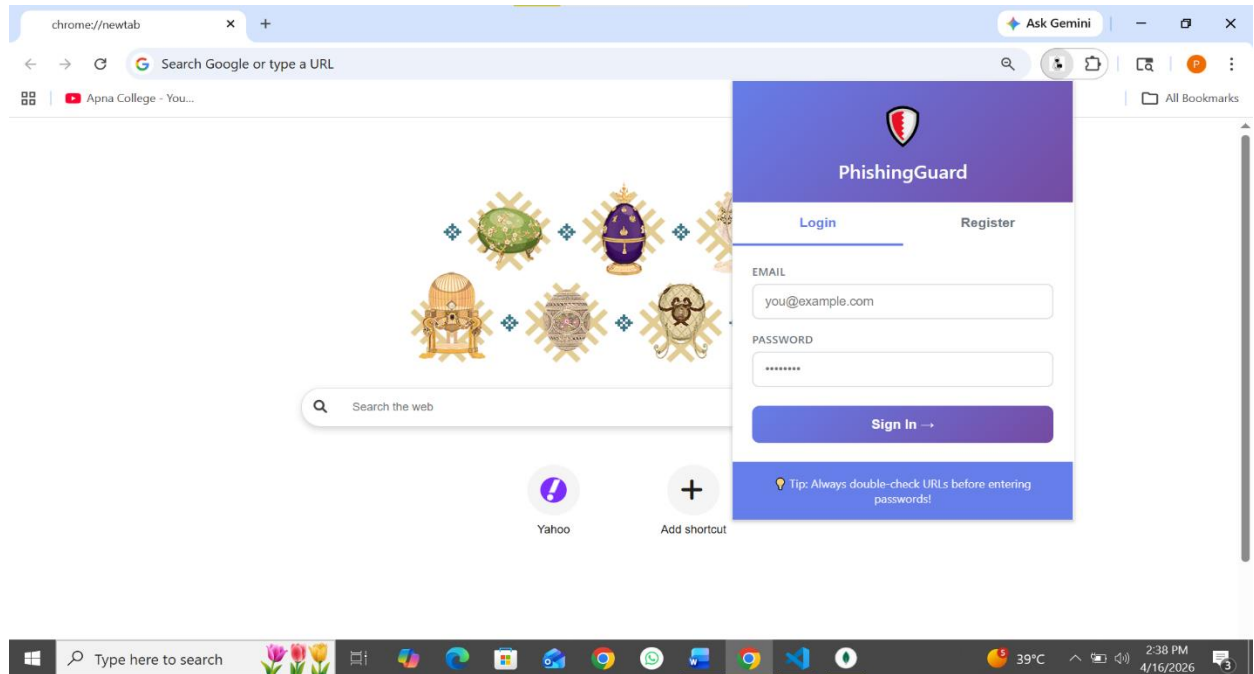
```
python-dotenv==1.0.0
```

Screenshots/ Images

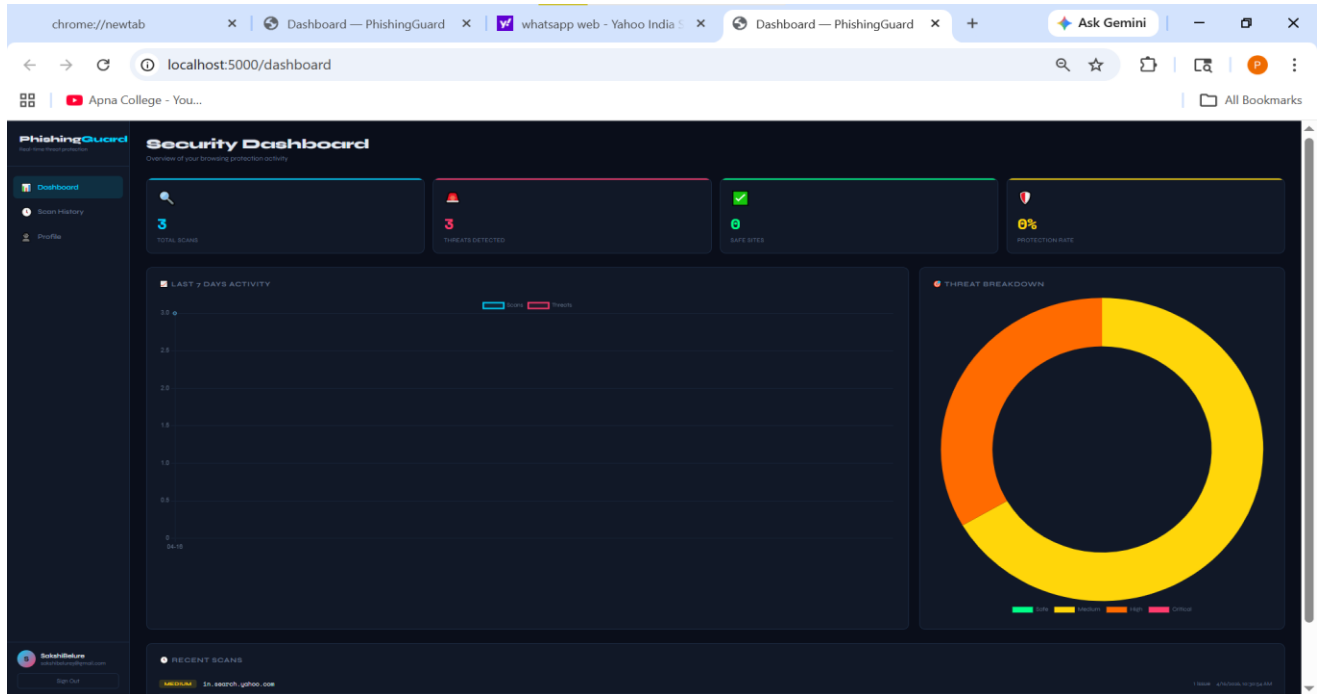
1.Registration:



2.Login:



3.Dadhboard:



4.History:

The screenshot shows the PhishingGuard Scan History page. It features a sidebar with navigation options: Dashboard, Scan History, and Profile. The main content area displays a table of scanned URLs with columns for STATUS, URL, THREATS, and TIME. The table lists three entries: two with a MEDIUM threat level for 'in.search.yahoo.com' and one with a HIGH threat level for '127.0.0.1'. Filter buttons for All, Safe, Medium, High, and Critical are located at the top of the table.

STATUS	URL	THREATS	TIME
MEDIUM	in.search.yahoo.com	1	4/16/2025, 10:30:34 AM
MEDIUM	in.search.yahoo.com	2	4/16/2025, 10:30:36 AM
HIGH	127.0.0.1	2	4/16/2025, 10:30:38 AM

5. Detected URL:

The screenshot shows the PhishingGuard Scan History page. The left sidebar contains links to Dashboard, Scan History, and Profile. The main content area displays a table of scanned URLs with their status (All, Safe, Medium, High, Critical). A detailed view of a scan for 'localhost' is shown on the right, indicating a 'Caution: Minor concerns detected' and a 'DETECTED ISSUES' section listing 'Site does not use HTTPS encryption'.

STATUS	URL
MEDIUM	localhost
MEDIUM	in.search.yahoo.com
MEDIUM	in.search.yahoo.com
HIGH	127.0.0.1

PhishingGuard
SakshiBelure

Caution: Minor concerns detected

CURRENT URL: localhost

DETECTED ISSUES:

- MEDIUM ⚠ Site does not use HTTPS encryption

▶ What do we check?

💡 Tip: Always double-check URLs before entering passwords!

6. User Profile:

The screenshot shows the PhishingGuard User Profile page. The left sidebar contains links to Dashboard, Scan History, and Profile. The main content area displays the user's profile information, including their name, email address, and member since date. It also shows statistics for total scans, threats blocked, and safe sites. Below this is a section for account information with fields for full name and email address, and an 'Edit Profile' button.

PhishingGuard
Real-time threat protection

Manage your account details

SakshiBelure
sakshibelure@gmail.com
Member since 4/16/2020

5 TOTAL SCANS
5 THREATS BLOCKED
0 SAFE SITES

ACCOUNT INFORMATION

FULL NAME
SakshiBelure

EMAIL ADDRESS
sakshibelure@gmail.com

Edit Profile

Future Scope

- **Integration of Advanced Machine Learning:**
Improve accuracy by using deep learning models to detect complex and new phishing attacks.
- **Real-time Threat Intelligence:**
Connect with online threat databases to update phishing detection rules automatically.
- **Automatic Website Blocking:**
Instead of only warning, the system can block access to malicious websites completely.
- **Multi-browser Support:**
Extend the extension to support browsers like Firefox, Safari, and others.
- **User Feedback System:**
Allow users to report suspicious websites, helping improve detection accuracy over time.
- **Cloud-based Analysis:**
Shift detection processing to the cloud for faster and more scalable performance.
- **Mobile Browser Support:**
Develop versions for mobile browsers to protect smartphone users.

Conclusion

This system enhances user security by reducing The “**Browser-Based Phishing Detection System (Extension)**” provides an effective solution to identify and prevent phishing attacks in real time. By analyzing URLs and webpage features, the system can classify websites as safe or suspicious and alert users instantly. The use of technologies like HTML, CSS, JavaScript, Python (Flask), Node.js, and MongoDB ensures smooth performance and efficient data handling.

the risk of data theft and improving awareness about phishing threats. Although it may have some limitations, such as false positives, it can be further improved with advanced machine learning techniques and real-time threat updates. Overall, the project demonstrates a practical and user-friendly approach to making web browsing safer.

References

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